A Lightweight Digital Twin for Thermal Behaviour Analysis of a Corridor Room

Using Low-Cost IoT Sensing, Occupancy Influence, and a First-Order RC Model

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Master’s thesis in Energy-efficient and Environmental Buildings

Lund University

Lund University, with eight faculties and a number of research centres and specialized institutes, is the largest establishment for research and higher education in Scandinavia. The main part of the University is situated in the small city of Lund, which has about 112 000 inhabitants. A number of departments for research and education are, however, located in Malmö. Lund University was founded in 1666 and has today a total staff of 6 000 employees and 47 000 students attending 280 degree programs and 2 300 subject courses offered by 63 departments.

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The degree project is the final part of the master's program leading to a Master of Science (120 credits) in Energy-efficient and Environmental Buildings.

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Abstract

This thesis investigates how much useful thermal information can be extracted from a small room using low-cost IoT sensors, simple physics, and no controlled experiments. A lightweight digital-twin framework was developed and deployed in an occupied student corridor room over a ten-day monitoring campaign. The sensing system, built from ESP32 nodes, a Raspberry Pi backend, FastAPI, and SQLite, operated continuously and produced clean, time-aligned measurements of indoor temperature, humidity, CO_2-equivalent, TVOC, PIR activity, and outdoor weather data. A rule-based fusion method combined PIR events with CO_2eq/TVOC trends to generate a stable binary occupancy signal without machine learning or labelled data.

A first-order 1R1C thermal model was calibrated using envelope properties and a free-float decay experiment, yielding , , and a time constant of . A single background heat-flux term captured unmodelled gains such as solar input and corridor spill-heat. Validation against 7,001 measured samples produced an overall RMSE of 0.92 °C, with 75.8% of timesteps within ±1 °C and no long-term drift in the residuals. Accuracy was higher during unoccupied periods (RMSE = 0.81 °C) than during occupied periods (RMSE = 1.15 °C), reflecting the limitations of the binary 104.4 W internal-gain assumption. A ±20% sensitivity analysis identified as the dominant parameter and confirmed numerical stability of the 120 s forward-Euler integration.

Short-term forecasting over 1–6 h horizons showed gradual RMSE degradation from approximately 0.95 °C to 1.35 °C, demonstrating that the room's long time constant preserves predictive accuracy well beyond the calibration window. These results show that a simple, interpretable, and computationally inexpensive model can reproduce the dominant thermal behaviour of a small room with practical accuracy, using only low-cost sensors and naturally occurring data.

The study contributes a reproducible monitoring stack, a transparent occupancy-fusion method, a practical calibration workflow, and empirical evidence that lightweight physics-based models remain viable for room-level digital twins. The findings suggest that setback-style control strategies, informed by the model and triggered by occupancy, could yield meaningful energy savings in thermally heavy spaces. Future work should explore multi-node RC models, improved occupancy estimation, integration with model-predictive control, and long-term deployments across full heating seasons

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Preface

This thesis was carried out between January 2026 and June 2026 as part of the master’s Program in Energy-Efficient and Environmental Building Design at Lund University. The work investigates how low-cost IoT sensing and simplified thermal modelling can be combined to create a lightweight digital twin for an existing building zone. The study was motivated by the growing need for affordable digital-twin solutions that can operate under real-world constraints, particularly in older buildings where high-quality sensor infrastructure is limited or absent.

The research was conducted independently by the author, including the design and development of the IoT sensing system, data collection, data processing, model development, and analysis. Guidance and feedback were provided throughout the project by the main supervisor, Pieter De Wilde, and the second supervisor, Ilia Larkov. Their support, technical insight, and constructive discussion were invaluable in shaping the direction and quality of this work.

I would like to express my sincere gratitude to DR. B. R. Ambedkar, the Government of Karnataka, and the Government of India for providing me with the opportunity and support to pursue my studies overseas. I am deeply thankful for the privilege. I also extend my heartfelt appreciation to my family and friends for their continuous encouragement, patience, and support throughout the thesis process.

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1 Introduction

Buildings account for approximately 30% of global final energy consumption, making their operational performance a critical factor in the transition toward sustainable built environments (IEA, 2023). In recent decades, digital twin technology has emerged as a transformative approach to building performance management, enabling real-time monitoring, predictive control, and data-driven operational decision-making. A digital twin functions as a continuously updated virtual replica of a physical system, synchronising sensor data with simulation models to support energy management, occupant-centric optimisation, and fault detection (Fuller et al., 2020; Boje et al., 2020). Recent systematic reviews indicate that digital twins can improve operational efficiency while identifying persistent challenges related to interoperability, sensing infrastructure, and data quality (Al Sehrawy et al., 2023). Despite this potential, adoption in existing buildings remains limited. Integration challenges, data-quality constraints, and the absence of standardised frameworks for physical-digital integration continue to restrict deployment (Boje et al., 2020; Al Sehrawy et al., 2023).

A central obstacle is the sensing infrastructure required to support a functional digital twin. Most implementations assume dense, high-quality sensor networks that are rarely available in the existing building stock (Khajavi et al., 2019). Retrofitting older buildings with such infrastructure is costly and operationally disruptive, limiting digital-twin deployment primarily to new or recently renovated buildings (Ma et al., 2012; Kineber et al., 2023). Low-cost Internet of Things (IoT) sensors offer a realistic alternative. Reviews of IoT-enabled digital twins show that such systems can improve indoor environmental quality and energy efficiency, but also highlight persistent issues with data reliability, sparse sensor setups, and environmental monitoring limitations (Al-Sehrawy et al., 2023). Al-Sehrawy et al. (2023) further argue that digital-twin implementations are not a single uniform technology but vary with the management aims behind them, distinguishing tech-driven, disruptive, cognitive, and humanistic approaches; the lightweight, sensor-constrained system built in this thesis sits closest to their tech-driven category, favouring a working low-cost implementation over an idealised, fully autonomous one. These challenges are particularly consequential in building-management contexts, where noisy and incomplete sensor streams must be interpreted carefully to extract actionable thermal and occupancy insights.

Lightweight digital-twin models, which use simplified low-order representations of building thermal dynamics, offer a practical pathway to bridge this gap. Such models operate in real time, require minimal computational resources, and remain physically interpretable, making them well-suited to small rooms and single-zone spaces where detailed simulation is neither warranted nor practical. Among these approaches, resistor–capacitor (RC) thermal networks have been established as an effective grey-box modelling method: they balance physical transparency with data-driven calibration and have been validated for heat-dynamics identification, model predictive control, and energy estimation (Bacher & Madsen, 2011; Foucquier et al., 2013). Recent reviews of digital twins for the built environment highlight increasing interest in simplified, interpretable models that support real-time monitoring and building operation (Fuller et al., 2020; Boje et al., 2020).

A persistent challenge in modelling small rooms is the influence of occupancy. Human presence introduces metabolic heat gains and alters indoor air composition through respiration and skin emissions, yet occupancy is difficult to measure directly without intrusive or expensive sensors. Low-cost alternatives, passive infrared motion detectors, CO_2-equivalent sensors, and total volatile organic compound sensors provide indirect signals that can be combined through sensor fusion to derive occupancy patterns (Yang & Becerik-Gerber, 2015; Chen et al., 2018). However, the integration of such simple occupancy-estimation methods directly into lightweight RC thermal models remains insufficiently explored. Most existing studies either assume known occupancy profiles or rely on smart sensing technologies that are impractical in low-cost deployments (Candanedo & Feldheim, 2016). Validation under naturally varying, uncontrolled conditions, where irregular occupancy, fluctuating outdoor temperatures, and sensor noise occur simultaneously, further limits the generalisability of existing findings.

This thesis develops and evaluates a lightweight digital twin of a small corridor room using low-cost IoT measurements and a first-order RC thermal model. The study focuses on modelling thermal behaviour using sparse sensor data while preserving physical interpretability and computational simplicity. Occupancy is estimated through a rule-based sensor-fusion approach combining PIR, CO_2eq, and TVOC measurements, and is incorporated into the RC model as a time-varying internal heat gain. The digital twin is calibrated and validated under naturally varying operating conditions representative of real building usage, without controlled experiments or artificially induced disturbances. The study also examines how the calibrated model can support occupancy-aware heating strategies and short-term temperature forecasting as a foundation for energy-efficient building operation.

1.1 Problem Statement

The introduction above framed this at building scale; concretely, this problem is visible at the scale of individual rooms, where many existing buildings operate without detailed knowledge of how indoor temperature responds to occupancy, outdoor conditions, and internal heat gains. Conventional modelling approaches frequently require extensive calibration procedures, high-quality sensing infrastructure, or controlled experimental conditions that are rarely feasible in occupied buildings. Consequently, building operations often depend on static schedules and simplified assumptions that do not accurately reflect real usage patterns.

There is therefore a need for a lightweight and interpretable modelling approach that:

- operates consistently with low-cost sensor data,
- functions under normal, uncontrolled building conditions,
- captures the dominant thermal dynamics of a single room, and
- provides insights that can support more energy-efficient operation.

This thesis addresses this need by developing a simplified digital twin of a corridor room using IoT measurements and a first-order RC thermal model.

A new, purpose-built model is developed here rather than reusing an existing tool (e.g. a whole-building simulation engine such as EnergyPlus, or a pre-calibrated model from another room) because such tools generally assume detailed geometric, material, and sensing inputs that are not available for this room, and because a model calibrated on a different space cannot be assumed to transfer to this one's specific envelope, occupancy pattern, and sensor placement. Developing and calibrating a dedicated, minimal-parameter model directly from this room's own low-cost sensor data avoids that transferability problem while remaining computationally cheap enough to run continuously on the same low-cost hardware used for data collection.

1.2 Research Gap

Although previous research has demonstrated the potential of digital twins, IoT monitoring systems, and RC thermal models in building applications, several limitations remain.

First, many digital twin studies rely on dense sensor networks and high-quality building management systems that are typically unavailable in older or retrofitted buildings. As a result, the applicability of such approaches to low-cost monitoring environments remains limited.

Second, relatively few studies investigate lightweight digital twins at the scale of individual rooms or small thermal zones, despite their practical importance for existing buildings with limited sensing infrastructure.

Third, while occupancy significantly influences indoor thermal behaviour, the integration of simple occupancy estimation methods into lightweight RC thermal models remains insufficiently explored. Existing studies often assume known occupancy profiles or rely on more advanced sensing technologies that are difficult to implement in practice.

Finally, many thermal modelling studies are validated using controlled experiments or artificially induced disturbances. Comparatively less attention has been given to model performance under naturally varying operating conditions representative of real building environments.

These limitations highlight the need for a lightweight digital twin capable of operating with sparse, low-cost IoT measurements while incorporating occupancy effects and maintaining reliable performance under real-world conditions.

This section restates, with literature grounding, the same four gaps introduced briefly in Section 1.2; Section 7.2 revisits each one directly against what this thesis actually delivers.

1.3 Research Questions

This study investigates how a lightweight digital twin can represent the thermal behaviour of a small corridor room using low-cost IoT measurements and a first-order RC thermal model. The research is guided by the following questions:

1. How accurately can a first-order RC model simulate indoor temperature using IoT sensor data collected from the corridor room?
2. How can occupancy be estimated using PIR, CO_2eq, and TVOC measurements, and how does occupancy influence the thermal behaviour of the room?
3. What thermal parameters and response characteristics can be identified through calibration of the RC model?
4. What energy-saving opportunities does the digital twin reveal, and how can these inform heating schedules or control strategies?

1.4 Aim and Objectives

This study aims to develop and evaluate a lightweight, sensor-driven digital twin of a small corridor room using IoT measurements and a first-order RC thermal model. The study investigates how simplified thermal modelling and occupancy estimation can represent indoor thermal behaviour under real operating conditions and support more energy-efficient building operation.

To achieve this aim, the study has the following objectives:

Methodological Objectives

- Establish an IoT-based monitoring system to collect indoor temperature, outdoor temperature, and occupancy-related measurements, including PIR, CO_2-eq, and TVOC data.
- Process and prepare the collected sensor data through time alignment, filtering, and sensor-fusion-based occupancy estimation.
- Develop a first-order RC thermal model representing the corridor room as a single thermal zone.
- Integrate occupancy-driven internal heat gains into the thermal model.
- Calibrate and validate the digital twin by comparing simulated and measured indoor temperature data under naturally varying operating conditions.

Analytical Objectives

- Examine the thermal behaviour of the room, including its response to occupancy patterns, outdoor temperature variations, and internal heat gains.
- Identify opportunities for improving energy-efficient operation, such as reducing heating during unoccupied periods and informing future predictive control strategies.

1.5 Scope and Limitations

This thesis focuses on developing a lightweight, sensor-driven digital twin of a small corridor room. It uses data from Internet of Things (IoT) devices and a first-order resistor-capacitor (RC) thermal model. The study is intentionally limited in scope to keep the model easy to understand, fast to compute, and practical for real-time building operations.

Scope:

- The study focuses on a single thermal zone, represented as a first-order RC system.
- Digital twin uses measured IoT data, including indoor temperature, outdoor temperature, PIR, CO2eq, and TVOC.
- Occupancy is estimated using a simple sensor-fusion method, and internal heat gains are calculated using a standard metabolic-rate value.
- The RC model is calibrated and validated under real operating conditions, without controlled experiments or artificial disturbance.

- The main aim is to understand the room's thermal behaviour, assess the accuracy of a lightweight digital twin, and identify energy-saving opportunities, such as reducing heating during unoccupied periods or enabling predictive control.

Limitations:

- The room is modelled as a single thermal mass, so spatial temperature variations and multi-zone effects are not captured.
- Only sensible heat gains from occupants are included. Equipment loads, lighting loads, and other internal heat gains are not considered.
- The IoT sensors are low-cost devices that generate data fluctuations and can compromise calibration accuracy.
- Factors such as infiltration, window leakage, and door-opening frequency are not modelled directly and are absorbed into the calibrated R and C values.
- The study does not implement or test a real energy-control strategy; it only identifies where savings are possible based on digital-twin insights.
- The results apply to this specific room and may not generalize to larger or more complex building zones without additional modelling.

The original study design included plug-load and lighting monitoring; these were excluded following pilot deployment due to sensor availability constraints, with the revised scope agreed with the thesis supervisors.

1.6 Thesis Structure

This thesis is organized into seven chapters that present the development and evaluation of a lightweight digital twin for a small corridor room using IoT measurements and RC thermal modelling.

Chapter 1 - Introduction

Introduces the motivation for monitoring indoor environments, outlines the research problem, and defines the objective. It also explains the role of the lightweight digital twin in understanding thermal behaviour and supporting energy-efficient building operation. The chapter concludes with an overview of the thesis structure.

Chapter 2 - Background

Reviews existing work on IoT-based environmental monitoring, occupancy detection, RC thermal modelling, and digital-twin applications. This chapter highlights the gap in lightweight, sensor-driven thermal twins for small rooms and motivates the modelling approach used in this study.

Chapter 3 - System Architecture and Experimental Setup

Describes the room selection, sensor selection, sensor node layout, and data-logging strategy. This chapter documents the real-world IoT system that provides the data foundation for the digital twin.

Chapter 4 - Data processing and Occupancy Modelling

Covers the cleaning and alignment of the raw sensor data, followed by the development of a focused occupancy signal using PIR and CO2eq measurements. The chapter concludes with the estimation of internal heat gain, which serves as a key input to the digital twin.

Chapter 5- RC Model Methodology

Presents the formulation of the first-order RC Model, including the continuous and stepwise equations, the integration of occupancy-driven internal heat gains, and the calibration of thermal parameters (R, C, k). This chapter forms the core of the digital twin, showing how real sensor data is transformed into a predictive thermal model.

Chapter 6 - Results and Discussion

Evaluates the performance of the digital twin by comparing simulated and measured indoor temperature. The chapter discusses what the model reveals about the room's thermal behaviour, how occupancy affects temperature, and how the calibrated model can support energy-saving strategies such as reducing heating during unoccupied periods or enabling predictive control.

Chapter 7 - Conclusion and Future Work

Summarizes the main findings, including the effectiveness of the lightweight digital twin, the accuracy of the RC model, and insights into thermal behaviour. The chapter also discusses how such a model can contribute to energy-efficient building operation and outlines opportunities for future research.

2 Background

2.1 Digital Twins for Building Operations

Digital twins first appeared in manufacturing, where engineers wanted a digital copy of a physical system that could be monitored and improved. The idea was first introduced by Michael Grieves (2006) during early work on product lifecycle management. Later, Grieves and Vickers (2017) described a digital twin as a physical object, a digital model, and the data connection that keeps them linked. Tao and Qi (2019) expanded this idea by explaining that a digital twin should update in real time and combine data from different sources so that the digital model always reflects what is happening in the real system.

In the building sector, digital twins are used for many purposes. They help monitor indoor conditions, detect faults in HVAC systems, support predictive control, and improve energy performance (Khajavi et al., 2019). Because they run continuously and use live data, digital twins can show how a building behaves at any moment, which makes them more useful than static simulation models.

However, using digital twins in existing buildings is still difficult. Many studies point out that most buildings do not have enough sensors to support a full digital twin (Boje et al., 2020). Retrofitting old buildings with new sensors and communication systems is expensive and often not practical (Rafsanjani et al., 2019). Another challenge is that many digital twins rely on detailed models that require a lot of calibration and computing power (Foucquier et al., 2013). These models are not always suitable for small rooms or simple applications.

Data quality is another issue. Low-cost sensors can drift, produce noise, or lose data, which affects the accuracy of the digital twin (Li et al., 2020). Because of these challenges, most digital-twin research focuses on large commercial buildings or new buildings with advanced systems (Khajavi et al., 2019). Small rooms and simple zones are rarely studied, even though they are easier to model and useful for testing lightweight approaches.

Researchers, therefore, call for lightweight digital twins, simple models that can run with low-cost sensors and still give useful insights (Fuller et al., 2020; Boje et al., 2020). These models should be easy to understand, fast to compute, and able to handle noisy data. For small rooms, simple thermal models like first-order RC networks can capture the main thermal behaviour without needing complex simulation tools (Foucquier et al., 2013).

This thesis follows that direction. It uses low-cost IoT sensors, a simple 1R1C thermal model, and a basic occupancy-estimation method. The model is calibrated using real data collected under normal conditions, without controlled experiments. The goal is to show that even a simple digital twin can help explain thermal behaviour and support energy-efficient operation in existing buildings. Many studies also assume access to high-quality sensor networks, which limits their use in older buildings. This thesis aims to show that a lightweight approach can still work with low-cost sensors and simple data.

A terminological caveat is warranted here. Common digital-twin taxonomies distinguish a digital model (no automated data flow), a digital shadow (automated one-way flow from the physical asset to its digital representation), and a full digital twin (automated bidirectional flow, where the digital representation can also act back on the physical system). The system built in this thesis ingests live sensor data and updates its state automatically, but it does not close the loop back onto the physical room: it does not control the heating or ventilation it monitors. By this stricter definition it functions as a digital shadow rather than a full digital twin, and the term "digital twin" is used throughout in its common, looser building-sector sense (as in Khajavi et al., 2019; Fuller et al., 2020) rather than as a claim of closed-loop control.

This also means the model cannot deliver energy savings by itself: a digital twin of this kind only identifies when and how heating could be reduced without loss of comfort. Realising those savings requires the model's output to be connected to an actuation layer, typically a building management system (BMS) or a simpler occupancy-triggered thermostat controller, that can act on the recommendation. This thesis does not implement that connection; it produces the room-level intelligence a BMS integration would consume, and treats that integration as necessary future work rather than an assumed capability of the model on its own.

The recurring reference to "older buildings" throughout this chapter is worth grounding concretely. A large share of the existing building stock in Europe, including much student and residential housing built during the mid-twentieth-century expansion, predates modern building-automation wiring, has masonry or concrete envelopes not designed around sensor or cabling routes, and cannot easily accommodate the dense, hard-wired sensor networks that conventional building-management and digital-twin systems assume. Retrofitting such buildings with that infrastructure is not simply expensive; it typically requires structural intervention (chasing walls for cabling, adding electrical capacity) that is disruptive to occupants and difficult to justify economically for a single room or a small residential unit (Ma et al., 2012; Kineber et al., 2023). Battery- or USB-powered wireless IoT nodes, of the kind used in this thesis, sidestep that structural barrier: they can be installed and removed without altering the building fabric, which is precisely why they are a realistic near-term option for this class of building rather than a full BMS retrofit.

2.2 IoT-Based Environmental Monitoring

IoT-based monitoring has become standard in buildings because it allows continuous measurement of indoor conditions at a low-cost. Many studies show that low-cost sensors can provide useful data for temperature, humidity, air quality, and occupancy-related signals (Al-Fuqaha et al., 2015). These systems are simpler to install than traditional building-management systems and are suitable for older buildings that lack advanced infrastructure.

Low-cost sensors, however, come with limitations. They often have lower accuracy, higher noise, and more drift over time compared to professional-grade instruments (Spinelle et al., 2017). For example, temperature sensors like the DHT series are sensitive to placement and airflow, and air-quality sensors such as CO_2-equivalent and TVOC sensors can be affected by humidity and warm-up time (Masson et al., 2015). Because of this, researchers usually apply filtering, calibration, or data-cleaning steps before using the measurements in models.

Microcontrollers like the ESP32 are widely used in research because they are cheap, support Wi-Fi, and can run simple firmware for data collection (Banzi & Shiloh, 2014). Many studies have used ESP32-based nodes to collect indoor temperature, CO_2, and motion data in practical environments (Al-Fuqaha et al., 2015). These systems are reliable enough for long-term monitoring when combined with a stable backend and proper data handling.

Air-quality sensors such as the ENS160 or similar TVOC/CO_2-eq sensors are often used to estimate occupancy trends. They do not measure true CO2 directly, but they respond to changes in human presence because people release CO2 and TVOCs when they breathe and move (Penza et al., 2020). Although these sensors are not as accurate as dedicated NDIR CO_2 sensors, they can still provide useful signals when combined with other data sources.

PIR sensors are another common tool in IoT monitoring. They detect motion by sensing changes in infrared radiation. PIR sensors are simple and reliable, but they only detect movement, not the number of people or how long they stay still (Yang & Becerik-Gerber, 2015). Because of this, PIR data is often combined with air-quality measurements to get a more complete picture of occupancy.

A major challenge in IoT monitoring is data quality. Missing data, communication drops, and sensor errors are common in real deployments (Li et al., 2020). This means that the backend system must handle timestamps, retries, and data validation. Many studies recommend using lightweight, serverless databases such as SQLite for small-scale deployments, since they are well-suited to embedded applications running on devices like the Raspberry Pi.

Existing studies confirm that low-cost IoT systems can provide sufficient data for basic digital-twin models, if the data is cleaned and interpreted carefully. These findings align with the method used in this thesis, where ESP32-based nodes collect temperature, air-quality, and motion data under normal building conditions. However, few studies evaluate how low-cost sensors perform during long-term deployment in naturally varying indoor environments, which is important for real-world digital-twin applications.

2.3 Occupancy Detection in Buildings

Knowing when people are present in a room is important for understanding how indoor temperature changes and for improving energy use. Many building-performance studies show that occupancy has a strong effect on heating, cooling, ventilation, and indoor air quality (Dong & Andrews, 2009). Therefore, researchers have explored different ways to detect or estimate occupancy using sensors that are easy to install and maintain.

One of the most common tools for detecting occupancy is the passive infrared (PIR) sensor. PIR sensors work by sensing changes in infrared radiation caused by human movement. They are affordable, simple, and widely used in lighting and HVAC control systems (Yang & Becerik-Gerber, 2015). Their main advantage is that they respond quickly when someone enters or moves in a room. However, PIR sensors also have clear limitations. They cannot detect people who are sitting still, and they cannot count how many people are present. They also have a limited field of view, which means that placement matters a lot. Because of these limitations, PIR data alone is often not enough for accurate occupancy estimation, especially in small rooms where people may remain still for long periods.

Another common method is to use indoor CO2 concentrations as an indicator of occupancy. Humans exhale CO2, so indoor levels tend to rise when people are present and fall when the room is empty. Several studies show that CO2 trends can be used to estimate occupancy patterns, especially in classrooms and offices (Penza et al., 2020; Jiang et al., 2021). CO2-based methods work well when occupancy changes slowly and when ventilation rates are known. But they also have drawbacks. CO2 sensors often have slow response times, and the concentration can lag behind actual occupancy changes. In small rooms with good ventilation, CO2 may not rise enough to give a clear signal. Low-cost CO2-equivalent sensors, which estimate CO2 from VOC readings, are even less accurate, but they can still show useful trends when combined with other data.

Total volatile organic compounds (TVOC) can also change when people are present because humans emit small amounts of VOCs through breathing, skin, and movement. Some studies have used VOC sensors to support occupancy estimation, especially when CO2 sensors are not available (Masson et al., 2015). VOC signals are usually noisy and influenced by many factors, but they can still help identify general occupancy patterns.

Because each sensor type has limitations, many researchers combine multiple signals to improve accuracy. Sensor-fusion methods often use PIR for first detection and CO2 or VOC trends for slower confirmation (Yang & Becerik-Gerber, 2015). This combination helps reduce false positives and false negatives. In practice, PIR detects movement and gives a quick indication of presence, while CO2eq or TVOC rising confirms that people stayed in the room. When both signals drop, the room is likely unoccupied. This approach remains computationally lightweight while maintaining sufficient accuracy for small-zone applications, especially in small rooms where occupancy changes are easy to track. More advanced studies use machine-learning methods to combine multiple sensor types, but these approaches require larger datasets and are harder to apply in real buildings (Jiang et al., 2021). For low-cost systems, simple rule-based fusion is often more practical.

The occupancy-estimation method used in this thesis follows the simple fusion approach found in the literature. PIR provides immediate detection of movement, while CO2eq and TVOC trends help confirm whether the room was occupied for a longer period. This method is suitable for low-cost sensors and works well in a small corridor room where occupancy is intermittent and easy to track.

Since occupancy directly influences internal heat gains, accurate occupancy estimation is essential for thermal modelling. The next section reviews RC models, which form the core of the thermal representation used in this thesis. Most fusion methods are evaluated in offices or classrooms, while small corridor-like spaces remain underexplored.

2.4 RC Models in Building Physics

RC models are widely used in building physics to describe how indoor temperature changes over time. They are based on a simple idea, heat flow in a building can be represented in the same way that electrical circuits represent the flow of current. In this analogy, thermal resistance (R) controls how easily heat moves through the building envelope, and thermal capacitance (C) represents how much heat the building materials can store. Because of this structure, RC models are often called “grey box” models, they are not fully detailed like simulation tools, but they still follow basic physical principles (Foucquier et al., 2013).

Different versions of RC models exist, depending on how much detail is needed. The simplest form is the 1R1C model, which treats the room as a single thermal mass with one resistance to the outside. More complex versions, such as 2R2C or higher-order networks, include additional resistances and capacitances to represent walls, internal layers, or furniture. These more detailed models can capture slower thermal effects, such as heat stored in thick walls, but they also require more parameters and more calibration (Fraisse et al., 2002). For small rooms or real-time applications, the simpler 1R1C model is often preferred because it is easier to work with and still captures the main behaviour of the space.

One of the main strengths of RC models is that they balance physical meaning with simplicity. Unlike black-box machine-learning models, RC models show how heat moves through the building, which makes it easier to interpret. They also run quickly and do not require a large dataset. This makes them suitable for digital-twin applications, where the model needs to update continuously based on sensor data. RC models can also be calibrated using real measurements, which help them reflect the actual behaviour of the room rather than depend only on theoretical material properties (Bacher & Madsen, 2011).

Estimating the values of R and C is an important part of using RC models. A common approach is to start with rough estimations based on material properties and then refine them using measured temperature data. Many studies use a “free float” period, when heating or cooling is off, to estimate the thermal capacitance from the natural cooling rate of the room. The thermal resistance is then adjusted by comparing the measured cooling behaviour with the expected behaviour from basic physical reasoning and data-based calculations. This combined approach is one of the reasons RC models work well in real buildings (Bacher & Madsen, 2011; Zhang et al., 2021).

RC models have been used in many digital-twin and building-control studies. Researchers have shown that simple RC models can support model-predictive control, energy estimation, and real-time monitoring, even when only low-cost sensors are available (Sturzenegger et al., 2016). Because they are lightweight and reliable, they are often recommended for small rooms, single-zone studies, and low-cost digital-twin applications.

In this thesis, the 1R1C model is selected because the corridor room is small, has simple geometry, and contains only one external wall, making it well-suited to treating the room as a single thermal mass. The available data originates from low-cost IoT sensors, which further supports the use of a simple and numerically stable model rather than a detailed simulation or a data-intensive machine learning approach. The RC formulation provides physically interpretable parameters, especially the thermal resistance (R) and thermal capacitance (C), which describe how the room responds to outdoor conditions and internal heat gains. These reasons make the 1R1C model a justifiable choice for developing a lightweight digital twin capable of operating consistently under normal building conditions.

This choice of model structure sits within a broader methodological choice common to building-performance modelling: whether to model the room from first-principles physics, as above, or to learn its behaviour statistically from data. The next section positions the RC approach within that wider landscape.

2.5 Data-Driven vs Physical-Based Modelling

Building-performance modelling generally follows two main approaches: physics-based models and data-driven models. Both are common in literature, and each offers its own advantages and limitations depending on the study's purpose, the amount of available data, and the required level of accuracy. Physics-based models use equations that describe heat transfer through walls, windows, and air. RC models, discussed earlier, belong to this category. More detailed physics-based tools, such as EnergyPlus, include multiple layers, materials, and boundary conditions to represent the building in greater detail (Crawley et al., 2001). These models are transparent because they follow known physical laws, and they remain valid even when data is limited. They are also useful for prediction and scenario testing. However, detailed physics-based models require many inputs, including material properties, infiltration rates, and internal loads. These inputs are often unknown in existing buildings, making calibration difficult (Foucquier et al., 2013). Simple models like the 1R1C approach avoid some of these limitations but may lose certain details.

Data-driven models take a different approach. Instead of using physical equations, they identify patterns directly from measurements with statistical or machine-learning methods. Common examples include linear regression, random forests, support vector machines, neural networks, and LSTM models for time-series prediction. These models can capture complex relationships without requiring detailed physical information, and numerous studies report that machine-learning methods can predict indoor temperature or energy use with high accuracy when sufficient data is available (Amasyali & El Gohary, 2018). Data-driven models also have drawbacks. They are often described as "black-box", which makes it difficult to interpret their internal workings. They may also perform poorly when conditions change, such as when occupancy patterns shift or when weather conditions differ from the training data. These limitations can be significant in low-cost IoT systems, where data may be incomplete or unreliable.

Some researchers combine both approaches to create hybrid models. Hybrid approaches combining physics-based and data-driven modelling have been proposed to overcome individual limitations by embedding physical laws into machine learning structures (Chen et al., 2025). Hybrid models can be powerful because they use the strengths of both methods, but they are more complex to design and require careful tuning. For this thesis, a simple physics-based model is more suitable than a data-driven or hybrid model. The corridor room is small and has simple thermal behaviour, and the available data comes from a low-cost sensor that may contain noise. The goal is to build a lightweight digital twin that can run in real time, so the model needs to be easy to interpret and explain. It should also function even when some data is missing or imperfect. RC models meet all these requirements. They are simple, fast, and reliable, and they provide meaningful parameters, thermal resistance, and thermal capacitance. These parameters help explain how the room behaves, which is important for understanding energy-efficient operation.

Existing studies confirm that both physics-based and data-driven models have value, but for small rooms with limited data, simple RC models are often the most practical choice. This finding supports the modelling approach used in this thesis.

2.6 Gaps in Current Research

Although there has been a lot of progress in digital-twin development, IoT-based monitoring, occupancy detection, and thermal modelling, several gaps remain in the research. Most digital-twin studies focus on large commercial buildings or new buildings that already have an advanced sensing system. Older buildings, which make up a large part of the building stock, often lack the sensors needed for high-quality digital-twin applications. Retrofitting these buildings with new equipment is expensive and not always practical, which limits the use of digital twins in real-world situations (Boje et al., 2020).

Another gap is the limited number of studies that explore lightweight digital twins for small rooms or single-zone spaces. Many existing digital-twin models rely on detailed simulations or machine-learning methods that require large datasets and high-quality measurements. These approaches are difficult to apply in small rooms with simple layouts and low-cost sensors. As a result, there is a need for simpler models that can still provide useful insights without requiring complex calibration or a large amount of data (Rasheed et al., 2020).

Occupancy detection is another area where gaps exist. While PIR sensors, CO_2 measurements, and VOC signals have all been used to estimate occupancy, very few studies combine these signals in a simple and practical way for thermal modelling. Most research either uses advanced machine-learning methods or depends on a single sensor type, which may not be reliable in all situations. There is limited work on integrating basic occupancy-fusion methods directly into thermal models, especially in the context of low-cost digital twins (Yang & Becerik-Gerber, 2015).

There is also a lack of studies that calibrate thermal models under normal operating conditions. Many calibration methods depend on controlled experiments, such as turning off heating or forcing specific temperature changes. These experiments are often not possible in occupied buildings. Only a small number of studies explore how to estimate thermal parameters like R and C using everyday sensor data collected during regular building use (Bacher & Madsen, 2011).

Finally, there is limited research on how lightweight digital twins can support energy-efficient operation in small rooms. Most digital-twin studies focus on large-scale energy management or advanced control strategies. Small rooms, however, also often offer opportunities for energy savings, especially when occupancy is intermittent. More research is needed to show how simple digital-twin models can identify these opportunities and support practical improvements (Khajavi et al., 2019).

These gaps highlight the need for a digital-twin approach that is simple, low-cost, and suitable for real-world conditions. This thesis addresses them by developing a lightweight digital twin for a small corridor room using low-cost IoT sensors, a basic occupancy-fusion method, and a first-order RC model calibrated under normal building operation.

Having identified this gap, the remainder of the thesis turns from literature to implementation: the next chapter describes the physical room, the IoT sensing system, and the experimental setup used to collect the data this model is built and tested on.

3 System Architecture and Experimental Setup

Data collection occurred between 1 and 11 April, with continuous measurements of indoor temperature, humidity, CO_2eq, TVOC, and PIR activity collected during this period.

3.1 Room Selection and Context

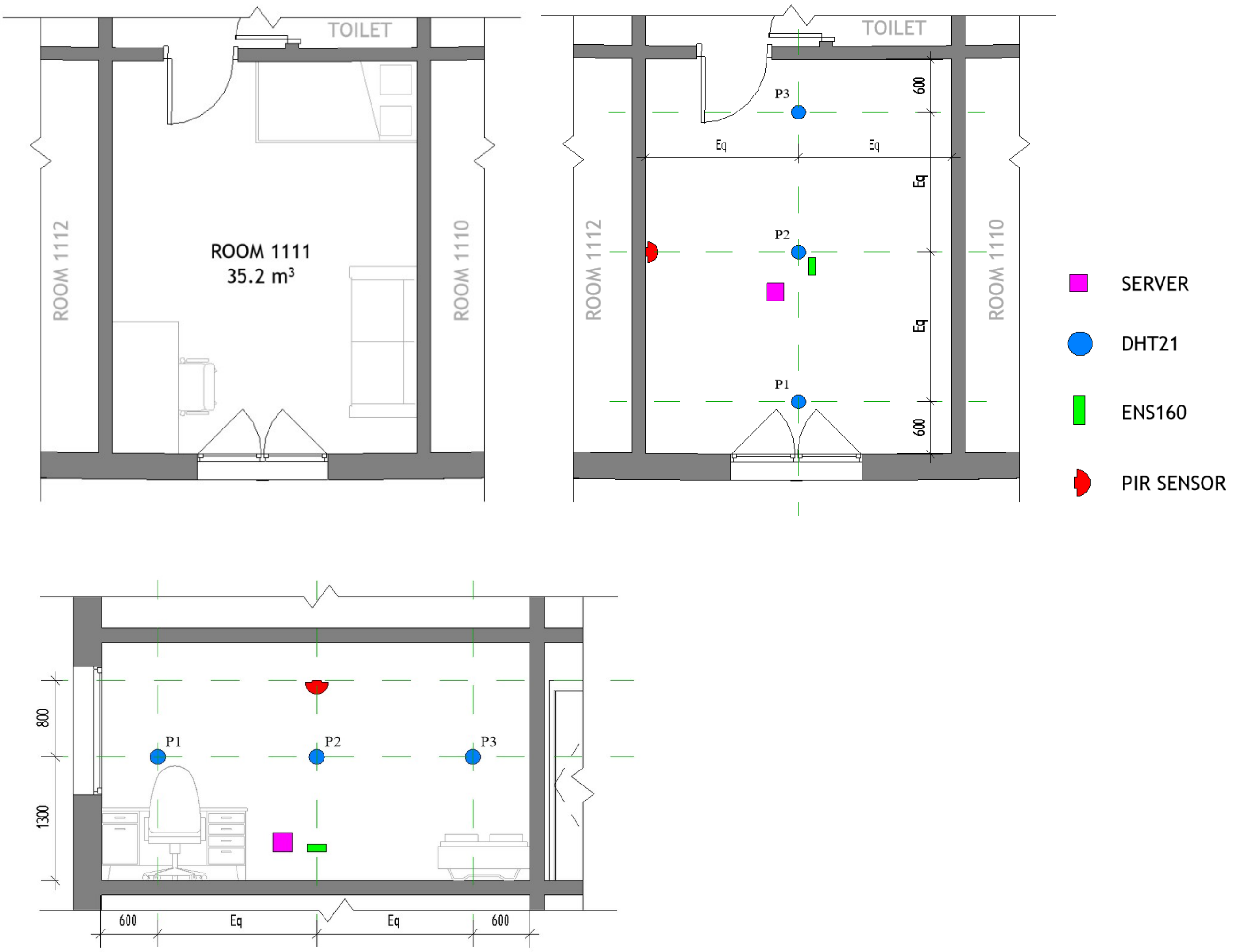
The monitored space selected for this study is a corridor room located on the first floor of the Parenthesen student housing complex in Lund, Sweden. Designed by architect Hans Westman, the Parenthesen complex was built in 1964, making it representative of the ageing Swedish student-housing stock this thesis is motivated by, rather than a modern, well-instrumented building. The room was chosen because it represents a small, lightweight space with clear external thermal exposure, and predictable occupancy patterns make it an ideal environment for evaluating the performance of a low-cost IoT-based digital twin.

The room contains one external wall with a single north-west-facing window, providing a clear and measurable thermal boundary. The remaining walls are internal partitions, which reduce the complexity of heat transfer pathways and support the use of a simplified 1R1C model. The space experiences intermittent occupancy throughout the day, allowing the study to examine how human presence influences indoor temperature through internal heat gains.

The room was also selected for practical reasons: It offered stable Wi-Fi coverage, accessible power outlets for the sensor nodes, and minimal disruption to occupants during installation. These conditions ensure reliable long-term data collection and support the deployment of the IoT sensing system under realistic building-operation conditions.

3.1.1 Room Description

Dimensions	Wall Construction	Window U-value	Ventilation Type	Heating System	Orientation
$3.5 \times 4.5 \times 2.5$ m ($V = 39.4$ m^3)	Plastered concrete (external) and plasterboard partitions (internal)	1.30 W/(m^2·K) (double glazed, 2.0 m^2)	Natural (grille closed during free-float)	Hydronic radiator (off during free-float)	North-west-facing exterior wall



Figur 1. Layout of ESP32 sensor nodes in the corridor room, showing DHT21 placement at 1.3 m, ENS160 air-quality sensor at 0.5 m, and PIR motion sensor at 2.0 m height.

This information defines the physical characteristics of the monitored space and provides the necessary context for interpreting the thermal behaviour captured by the digital twin.

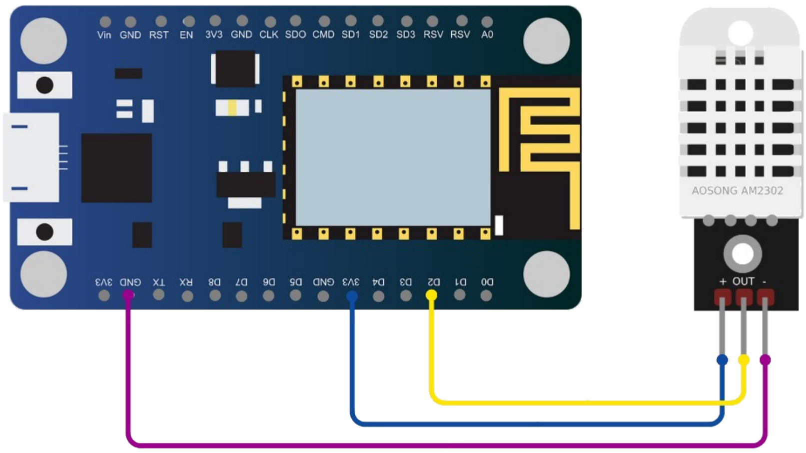
3.2 Overview of the IoT Sensing System

The IoT sensing system was designed to collect indoor environmental and occupancy-related data from the corridor room. The system consists of multiple ESP32-based sensor nodes connected to a Raspberry Pi server over the local Wi-Fi network. Each node measures a specific variable, such as temperature, air quality, or motion, and transmits the data to the server every 2-seconds, providing high-resolution measurements and ensuring that short-term fluctuations are captured. During data cleaning and sensor fusion processing, the raw 2-second readings were resampled to a 2-minute interval to create a consistent timeline for analysis and to match the timestep used in the RC thermal model. The system operates continuously under normal building conditions, without controlled experiments or artificial disturbances, ensuring that the collected dataset reflects real-world behaviour and is suitable for developing and validating both the digital twin and the RC model.

3.3 Hardware Components

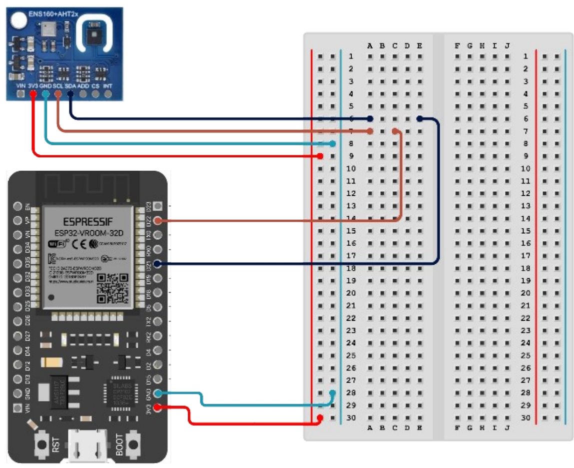
The sensing system uses low-cost, commercially available components. Each sensor node includes an ESP32 microcontroller and one or more sensors, depending on the measurement type. The following subsections describe the system's main hardware components.

3.3.1 Temperature Sensors (DHT21)



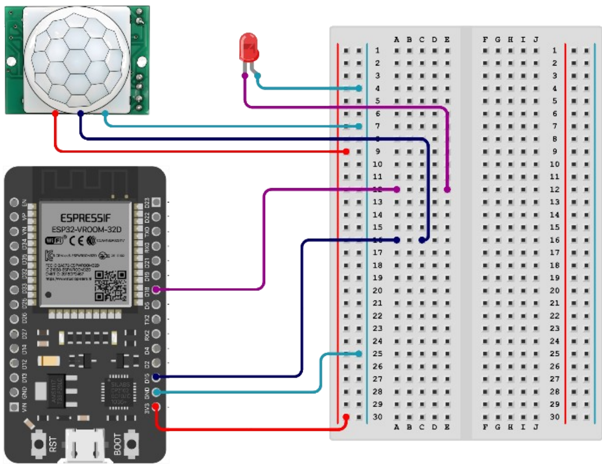
The DHT21 sensor is used to measure indoor air temperature at multiple points throughout the corridor. It provides digital output and has an accuracy of ±0.5°C under typical indoor conditions. The sensors were placed at a height of 1.3m and distributed horizontally to capture spatial variation. The DHT21 was selected due to its low-cost, simple interface, and suitability for long-term indoor monitoring.

3.3.2 ENS160 Air-Quality Sensor



The ENS160 sensor measures air-quality indicators such as CO2-equivalent (CO2eq) and total volatile organic compounds (TVOC). These signals are used to support occupancy estimation by capturing changes in indoor air composition. The sensor was placed near the center of the room at a height of 0.5 m. Environmental compensation using temperature and humidity readings from the DHT21 was applied in the firmware to improve measurement accuracy.

3.3.3 PIR Motion Sensor

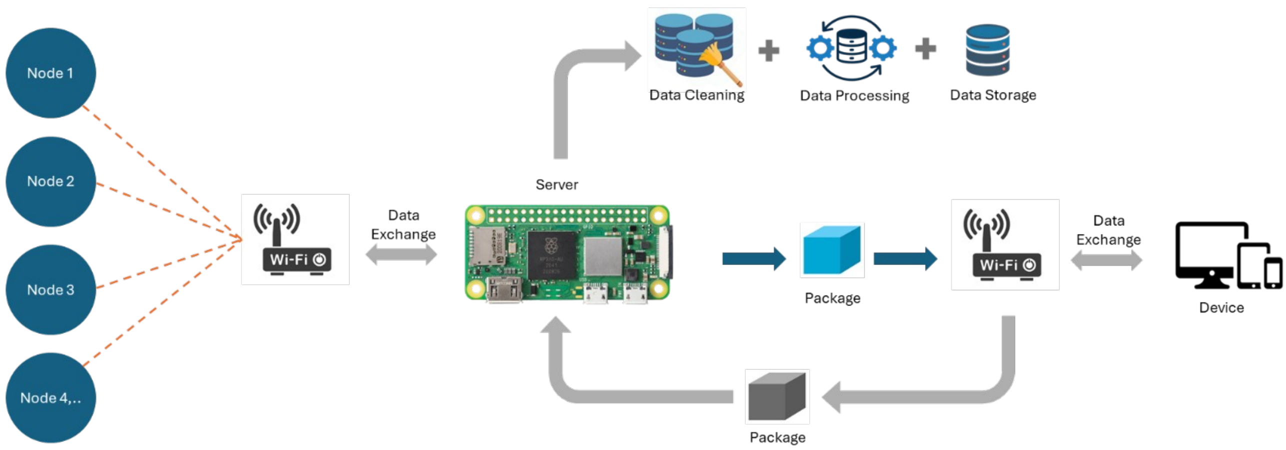


A passive infrared (PIR) sensor is used to detect human movement in the corridor room. The sensor was mounted on the wall at a height of 2.0 m to provide a clear field of view. The PIR output is binary, indicating whether motion is detected within the sensor's range. This signal is used in the occupancy-estimation model.

3.3.4 ESP32 Microcontroller

The ESP32 microcontroller serves as the core of each sensor node. It provides Wi-Fi connectivity, processes sensor readings, and transmits data to the Raspberry Pi server. The ESP32 was chosen for its low power consumption, built-in wireless communication, and compatibility with the sensors used in this study. Each node runs a simple firmware loop that reads sensor values, applies basic filtering, and sends the data at a regular interval.

3.4 Firmware and Data Acquisition



Each sensor node runs a simple firmware program on the ESP32 microcontroller. The firmware is responsible for reading sensor values, organizing the data, and sending it to the backend system. It acts as the interface between the physical sensors and the digital processing pipeline.

When the device starts, the firmware initializes all connected sensors and checks that they are ready for operation. After initialization, the firmware follows a fixed sampling interval to ensure consistent data collection. Temperature, CO2eq, TVOC, and PIR readings are obtained locally and time-stamped by the ESP32.

The firmware performs basic filtering, such as rejecting negative values or sensor warm-up spikes, to remove obvious noise and apply simple checks to avoid invalid, missing, and out-of-range values. All readings are stored temporarily in buffers until a stable Wi-Fi connection is available. Once communication is established, the firmware packages the data into a JSON structure and sends it to the Raspberry Pi server using HTTP POST requests. If the connection fails, the firmware retries until the data is successfully delivered.

This setup ensures continuous and reliable data acquisition under normal building conditions without manual intervention.

3.5 Backend Architecture (FastAPI, SQLite, Nginx)

This backend system runs on a Raspberry Pi Zero 2 W and receives data from all sensor nodes. It consists of three main components:

FastAPI

FastAPI handles incoming HTTP requests from the ESP32 nodes. It validates the JSON payloads, adds a server-side timestamp, and forwards the data to the database. FastAPI was selected specifically because the RC model in Chapter 5 is calibrated and run on a fixed 120-second timestep: a lightweight, low-latency framework was needed so that concurrent requests from multiple ESP32 nodes are timestamped and stored without queuing delay, which would otherwise introduce timing jitter into the very series the model treats as evenly spaced.

SQLite

All validated sensor readings are stored in a local SQLite database. SQLite is used because the ten-day calibration and validation dataset in Chapter 6 depends on a single, gap-free, correctly ordered log: SQLite requires no external server that could itself fail or need maintenance during the unattended monitoring campaign, and its structured queries make it straightforward to extract and resample the 2-second raw readings into the 2-minute intervals the RC model consumes.

Nginx

Nginx acts as a reverse proxy in front of FastAPI. It manages incoming traffic, improves stability, and ensures that the backend remains responsive even when multiple nodes send data at the same time.

Together, these components form a stable and low-cost backend suitable for continuous indoor monitoring.

3.6 Data Logging and Storage

All sensor data is stored in the SQLite database on the Raspberry Pi. Each entry includes:

- node ID
- sensor type
- measured value
- ESP32 timestamp
- Server timestamp

Server timestamps are generated by the Raspberry Pi system clock. The database grows over time, so periodic maintenance is performed, including log rotation and database vacuuming to reduce file size and maintain database performance. Daily backups are stored locally to prevent data loss. This logging approach provides a complete dataset for later processing, occupancy estimation, and RC model calibration.

3.7 Setting up in the Corridor Room

The sensor nodes were installed in the corridor room under normal operating conditions.
Placements were chosen to capture representative measurements:

- Temperature sensors were placed at a height of 1.3 m.
- The ENS160 air-quality sensor was placed at a height of 0.5 m.
- The PIR sensor was mounted at 2.0 m height for a clear field of view.
- The Raspberry Pi server was placed outside the main walking path to avoid interference.

No artificial disturbance or controlled experiments were introduced. The system operates continuously for the full monitoring period, providing real-world data for the digital twin.

3.8 Ventilation Condition

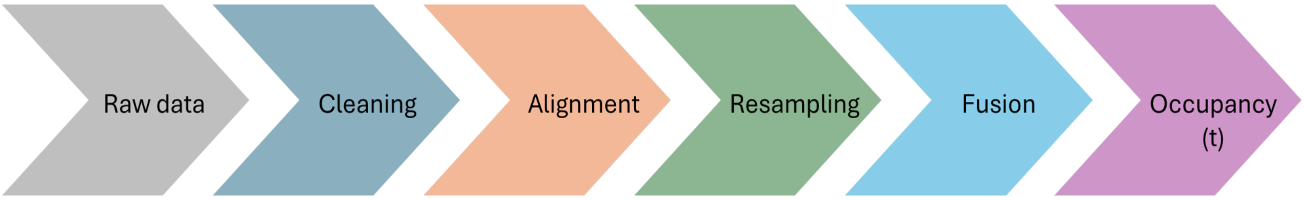
During the free-float measurement period, the ventilation grille was fully covered to eliminate air-exchange effects. This ensured that the cooling behaviour reflected only the thermal envelope and internal mass, improving the reliability of RC parameter identification.

3.9 Ethical and Privacy Considerations

The sensing system does not record personal data. The PIR sensor detects motion but does not identify individuals. CO2eq and TVOC measurements indicate air-quality changes, but cannot be linked to specific people. No audio, video, or image data is collected. All data is stored locally on the Raspberry Pi and is used only for research purposes. The deployment follows standard ethical guidelines for indoor environmental monitoring and respects occupant privacy.

With the sensing hardware and data-collection protocol in place, the next chapter turns to what is done with the raw measurements: cleaning, aligning, and fusing them into the occupancy signal and model inputs used throughout the rest of the thesis.

4 Data Processing and Occupancy Modelling



4.1 Raw Data Overview

The raw dataset consists of multiple environmental and occupancy-related variables collected from the corridor room under normal operating conditions. These include indoor temperature, outdoor temperature, CO_2-equivalent (CO_2eq), total volatile organic compounds (TVOC), and PIR motion detections. All sensors recorded data continuously, with each ESP32 node sampling at 2-second intervals before the data were aligned and resampled for analysis.

The raw data displays typical issues associated with low-cost, real-world sensing systems, including noise, missing values, timestamp drift between nodes, and occasional sensor warm-up artefacts. These irregularities must be addressed before the dataset can be used for occupancy estimation or RC-model calibration. Subsequent sections describe the cleaning, synchronization, and fusion steps applied to transform the raw sensor streams into a consistent, high-quality dataset suitable for thermal modelling and digital-twin development.

4.2 Pre-Processing and Cleaning

The raw sensor data contain noise, missing values, timestamp irregularities, and occasional instability during sensor warm-up that must be addressed before the dataset can be used for occupancy estimation or RC-model calibration. The pre-processing stage removes invalid readings, reduces noise, and prepares each variable for later fusion and modelling. Invalid measurements are discarded before analysis, including temperature values outside the DHT21 operating range (−40 °C to 80 °C), sudden spikes in CO_2eq or TVOC exceeding three standard deviations from a rolling mean, consecutive PIR triggers within 5 seconds (treated as electrical noise or false activation), and warm-up readings recorded immediately after sensor reboot. This ensures that only physically meaningful data is retained.

To reduce short-term fluctuations while preserving overall trends, a moving average filter with a window of five samples is applied to the temperature, CO_2eq, and TVOC signals. This smoothing is performed after resampling to the fixed 2-minute interval, so the five-sample window corresponds to 600 seconds. The smoothing step suppresses isolated spikes and improves the stability of the dataset. PIR data remain unsmoothed to preserve the sharp transitions required for occupancy detection. Gaps of 10 minutes or less are filled using linear interpolation, while gaps exceeding 10 minutes are left unfilled and excluded from calibration and occupancy fusion. After filtering, smoothing, and removal of invalid values, the cleaned sensor streams form a consistent and reliable dataset suitable for timestamp alignment, resampling, and the occupancy fusion process described in the following sections.

4.3 Time Alignment and Resampling

Because the sensor nodes operate independently, their internal clocks drift slightly over time, resulting in small timestamp mismatches between data streams. All measurements are first aligned to a common timeline using the server timestamp as the reference. The dataset is then resampled to a fixed 2-minute interval, matching the time-step formulation of the RC model. Short gaps of up to 10 minutes are filled using linear interpolation to maintain continuity, while longer gaps are left unfilled and excluded from model calibration to avoid introducing artificial trends. This process produces a synchronised and uniformly sampled dataset suitable for occupancy detection and thermal modelling.

4.4 Sensor Fusion for Occupancy Detection

Occupancy in the corridor room is estimated by combining information from PIR motion events and air quality trends. Each sensor captures a different aspect of human presence: PIR detects immediate movement, while CO_2eq and TVOC reflect slower, sustained changes caused by metabolic emissions. By integrating these complementary signals, the fusion method improves detection accuracy and reduces false positives compared to using either sensor alone.

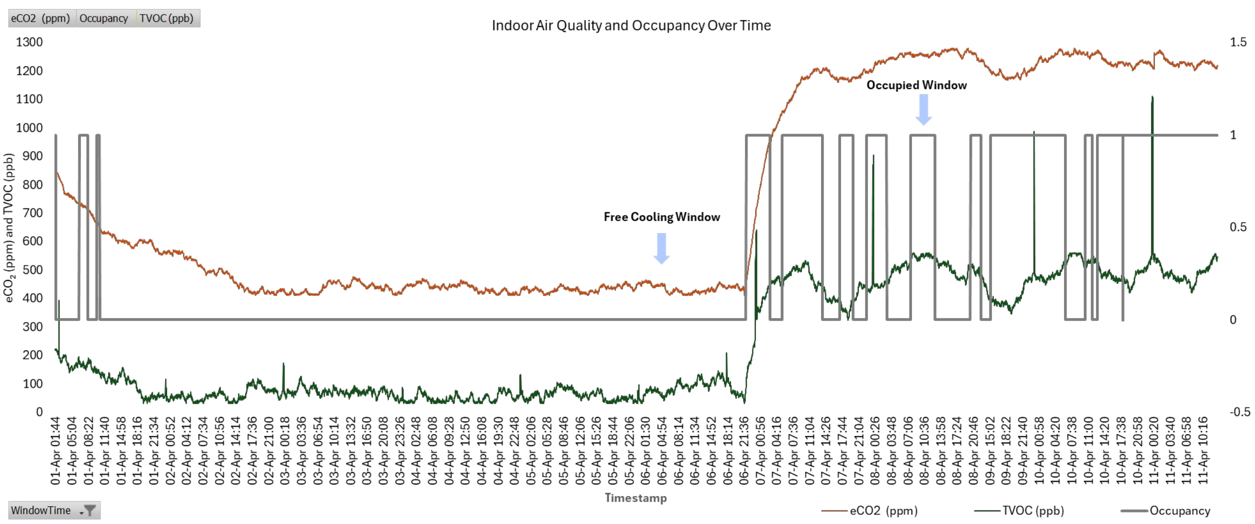
4.4.1 PIR Motion Events

The PIR sensor provides rapid detection of human movement and responds immediately when a person enters or moves within the room. This makes it particularly effective for identifying short visits, transitions, and brief occupancy periods that may not produce noticeable changes in air quality. However, PIR cannot detect stationary occupants and may miss longer periods of presence when no movement occurs. For this reason, PIR alone is insufficient for reliable occupancy estimation in small, intermittently used spaces.

4.4.2 CO_2eq and TVOC Trends

CO_2-equivalent (CO_2eq) and total volatile organic compound (TVOC) levels increase when people are present due to human respiration and metabolic emissions. Occupants exhale CO_2 continuously, while skin emissions and personal care products release VOCs into the indoor environment. These behaviours are well-established in indoor-air-quality research and are commonly used as indicators of human presence (Penza et al., 2020; Masson et al., 2015). Once the room becomes unoccupied, fresh air gradually dilutes these contaminants, causing concentrations to decline towards baseline levels.

In the small corridor room studied here, typical baseline CO_2eq levels range from 400-500 ppm when unoccupied. During occupancy, CO_2eq rises detectably, with sustained increases of 20 ppm or more over 10 minutes, indicating likely presence. TVOC levels follow a similar pattern, although they are more variable due to external influences such as cleaning products, building materials, or transient odours. These slower-changing air-quality signals complement the fast response of PIR sensors and provide reliable information about sustained occupancy, consistent with findings in previous occupancy-modelling studies (Yang & Becerik-Gerber, 2015; Jiang et al., 2021).



Figur 2. CO_2eq, TVOC, and PIR activity over a representative 24-hour period, used to derive the binary Occupancy(t) signal via rule-based fusion.

Table: Expected CO_2-eq and TVOC Changes with Occupancy

Occupancy State	CO_2eq (ppm)	TVOC (ppb)	Typical Duration	Observed Trend
Unoccupied (baseline)	400-500	<100	Extended periods	Stable, low values

Occupant enters	420-550	100-200	First 1-5 minutes	Gradual rise begins
Sustained occupancy (1 person)	550-800	150-400	15–60 minutes	Steady increase, then plateau
Prolonged occupancy (1+ hours)	800-1200	300-600	>60 minutes	Slow continued rise or stable elevated level
Occupant leaves	500-700	150-300	First 5–15 minutes	Gradual decline
Return to baseline	400-500	<100	20–60 minutes	Exponential decay toward ambient

These values are representative of a small corridor room (approximately 20–30 m³) with natural ventilation.. Actual concentrations depend on room volume, air-exchange rate, number of occupants, and sensor accuracy. The thresholds used in the fusion logic, such as a 20 ppm rise over 10 minutes or a 15 ppm decline towards baseline, are derived from the observed behaviour in this specific room.

4.4.3 Fusion Logic

The fusion logic combines the fast-response PIR signal with the slower but more stable CO_2eq and TVOC trends to produce a reliable occupancy estimate. Each sensor captures a different aspect of human presence: PIR detects immediate movement, while air-quality indicators reflect sustained occupancy. By integrating these complementary signals, the method reduces false positives and false negatives that would occur if either sensor type were used alone.

The fusion operates on two core principles:

1. **Detecting the onset of occupancy**

Occupancy is set to 1 when either of the following conditions is met:

PIR detects motion, indicating that someone has entered or moved within the room.
CO_2eq or TVOC levels show a sustained rise, typically a ≥20 ppm increase in CO_2eq over 10 minutes, signalling that a person has been present long enough to influence indoor air composition.

This ensures that both short visits and longer stays are captured, even if the occupant remains still after entering the room.

2. **Detecting the end of occupancy**

Occupancy returns to 0 when:

PIR remains inactive for an extended period, and
CO_2eq and TVOC levels decline toward baseline, typically a ≥15 ppm decreases in CO_2eq over 10–15 minutes.

This prevents false occupancy detection caused by residual air-quality elevations after the occupant has left.

Outcome of the fusion

The resulting binary signal, Occupancy(t), is stable, interpretable, and aligned with the resampled 2-minute dataset. It captures both rapid transitions and sustained presence, making it suitable for modelling internal heat gains and analysing how occupancy influences the thermal behaviour of the corridor room.

4.5 Construction of Occupancy(t)

The final occupancy signal, Occupancy(t), is constructed by applying the fusion logic described in the previous section to the resampled 2-minute dataset. The PIR, CO_2eq, and TVOC signals are first aligned to the common timeline, after which the rule-based fusion method determines whether the room is considered occupied at each timestep.

The resulting Occupancy(t) is a binary time series, where:

- Occupancy(t) = 1 indicates that the fusion logic has detected either recent motion or a sustained rise in air-quality indicators consistent with human presence.
- Occupancy(t) = 0 indicates that no motion has been detected and CO_2eq/TVOC levels are declining towards baseline.

This representation captures both short visits and longer periods of presence, while filtering out noise and transient fluctuations. Because the signal is aligned with the RC model's time-step formulation, it can be used directly to compute internal heat gains and to analyse how occupancy influences the thermal behaviour of the corridor room.

4.6 Internal Heat Gain Modelling

Internal heat gains arise from the metabolic activity of occupants and act as an additional heat source within the thermal balance of the room. To represent this effect, a standard metabolic rate of 1 MET, equivalent to 58 W/m², is used. Assuming an average adult body surface area of 1.8 m², the sensible heat contribution per person is:

$$Q_{\mathbf{person}} = 58 \times 1.8 = 104.4\ \mathrm{W}$$

Only sensible heat is considered in this study, as latent heat contributions are relatively small in this space and have limited influence on the thermal response of the corridor room.

The binary occupancy signal , derived from the fusion of PIR, CO_2eq, and TVOC data, indicates whether the room is occupied at each timestep. Internal heat gains are therefore computed as:

When the room is occupied, Occupancy(t) = 1, and the model adds 104.4 W of internal heat. When the room is unoccupied, Occupancy(t) = 0, and the internal heat gain is zero. This formulation provides a simple and computationally efficient way to incorporate human presence into the RC model and ensures that the simulated indoor temperature reflects the warming effect of occupancy.

Tabell 1. Parameters Used for Internal Heat Gain Calculation

Parameter	Value	Description
Metabolic rate	1 MET	Standard light-activity metabolic level
Sensible heat per unit area	58 W/m²	Heat output associated with 1 MET
Average adult body surface area	1.8 m2	Typical value for an adult occupant
Sensible heat per person	104.4 W/person	Computed as 58 x 1.8

5 RC Model Methodology

The thermal behaviour of the corridor room is represented using a first-order resistor and capacitor (1R1C) model. This chapter describes the formulation of the model, the discretization used for simulation, the incorporation of occupancy-driven internal heat gains, and the calibration of the thermal parameters. A backward-looking consistency check is performed using a free-cooling period to refine the thermal capacitance. The chapter concludes with the validation of the calibrated model and its application for forward temperature forecasting.

As established in Section 2.4, a first-order 1R1C representation is adopted here, rather than a higher-order (e.g. 2R2C) or fully physical model, because the corridor room has a single external wall, simple geometry, and only two dominant heat-exchange pathways: envelope conduction and occupancy-driven internal gains. A higher-order model would introduce additional parameters that cannot be reliably identified from the sparse, low-cost sensor data available in this study, whereas the 1R1C form keeps every parameter physically interpretable and estimable from the available measurements.

5.1 Continuous 1R1C RC Model Formulation

The 1R1C model represents the room as a single thermal zone with aggregated thermal resistance and capacitance. The standard first-order energy-balance equation:

(1)

Where is the indoor air temperature, is the outdoor air temperature, R is the thermal resistance between the indoor and outdoor environments, C is the thermal capacitance of the room, and denotes internal heat gains. The term describes heat transfer through the building envelope, while Q_{int} corresponds to heat generated inside the room. This lumped-parameter, single-node formulation follows the standard grey-box RC thermal-network representation established in building physics (Fraisse et al., 2002; Bacher & Madsen, 2011).

In this formulation, the indoor air temperature changes according to heat transfer through the envelope and internal heat gains. The model assumes a well-mixed indoor air volume, constant thermal properties, negligible latent heat effects, and no explicit infiltration modelling. These simplifications are reasonable for lightweight digital-twin applications and allow the model to operate efficiently with sparse IoT data.

5.2 Stepwise Model

To simulate the model using time-series sensor data, the continuous equation is discretized using a forward-Euler approximation. The indoor temperature at the next timestep is computed as:

(2)

In this formulation, represents the indoor temperature at time , which can be either the most recent sensor reading or a previous model prediction, depending on the simulation mode. The outdoor temperature is obtained from weather forecasts or measured values, while the internal heat gains are derived from the occupancy signal . A fixed timestep of 120 seconds is used, matching the IoT logging interval. This discrete formulation is applied consistently for calibration, backward-looking consistency checks, validation, and forecasting. The same equation applies across all simulation modes; only the input sources differ.

5.3 Internal Heat Gain from Occupancy

Occupancy contributes to internal heat gains through metabolic activity. A standard metabolic-rate assumption of 104.4 W per person is used. The binary occupancy signal Occ(t), derived from the sensor-fusion method in Chapter 4, is used to compute internal heat gains:

$$Q_{\mathrm{int}}(t) = 104.4 \times \mathrm{Occ}(t)$$

This ensures that the model responds dynamically to occupancy-driven temperature changes.

5.4 Parameter Calibration

5.4.1 Estimating Thermal Resistance (R)

The thermal resistance *R* gives the overall heat-loss path between the indoor air and the outdoor environment. It is estimated to use the material properties and areas of the room envelope according to:

Using the measured values (the 8.75 m² external-wall area below is net of the 2.0 m² window opening, i.e. it excludes the window; the window is accounted for separately at its own U-value):

Surface	Area (m²)	U-value (W/m²K)
External wall	8.75	0.227
Window	2.00	1.30

The overall heat-loss coefficient becomes:

$$UA = (0.227)\ (8.75) + (1.3)\ (2.0) = 4.58625 \text{ W/K}$$

The corresponding thermal resistance is:

$$= 0.218 \text{ K/W}$$

This value serves as the initial estimate for the RC model.

5.4.2 Calibrating Thermal Capacitance (C) Using Free-Cooling

A free-float cooling period was used to estimate the thermal capacitance . During this period, heating was off, the ventilation grille was fully covered, no occupants were present, and no internal heat gains occurred. Under these conditions, the RC model simplifies to:

Using the previously calibrated thermal resistance . The free-float dataset provided a clean temperature-decay signal. The indoor temperature decreased from 19.59 °C to 19.47 °C over 7 200 seconds (120 minutes), while the outdoor temperature remained at 5.15 °C. The temperature change of -0.12 °C to a decay rate of:

The average indoor temperature during the decay was **19.53 °C**, giving a driving temperature difference of:

Substituting these values into the simplified RC equation yields:

The negative signs cancel, giving a positive thermal capacitance:

This value represents the combined thermal mass of the air, furniture, and internal surfaces.

5.4.3 Calibrating Background Heat Flux ()

After estimating from envelope properties and From the free-float decay, a systematic offset remained between simulated and measured indoor temperature during unoccupied periods. This offset arises from heat sources not explicitly represented in the 1R1C model: solar gain through the north-west-facing window, conductive heat transfer from warmer neighbouring rooms through internal partitions, and slow heat release from deep thermal mass layers.

To compensate, a constant background heat flux was introduced into the energy balance:

Qbg was estimated by minimising the RMSE between simulated and measured indoor temperature over the full unoccupied subset of the validation dataset (5,061 timesteps). A scalar grid search was performed over the range 0 to 150 W in increments of 0.5 W, with and held fixed at their calibrated values and throughout. The value produced the minimum unoccupied RMSE and was adopted as the calibrated parameter.

This calibration procedure means the unoccupied-period accuracy figure reported in Chapter 6 is not a fully independent validation statistic: Q_{bg} was tuned specifically to minimise RMSE on the same 5,061-timestep unoccupied subset that this figure describes, so the reported unoccupied RMSE is close to the calibration objective's own residual rather than an out-of-sample test of it. R and C, by contrast, were calibrated separately from the free-float window and are validated out-of-sample against this subset. This distinction should be kept in mind when interpreting the closeness of the unoccupied fit as evidence of model accuracy.

5.5 Backward-Looking Consistency Check of *R* and *C*

After estimating and computing a backward-looking consistency check was performed to ensure that the model reproduced the observed free-cooling behaviour. The model was simulated over the entire free-float interval using measured outdoor temperature, zero internal gains, and the measured indoor temperature at the start of the decay. If the model cooled too quickly, the effective thermal capacitance was increased to represent a larger thermal mass. If the model cooled too slowly, the capacitance was reduced. This iterative refinement continued until the simulated and measured cooling curves aligned in both shape and magnitude. The final values of and were therefore both physically plausible and empirically consistent.

5.6 Model Validation

The calibrated RC model was validated using a separate dataset collected under normal operating conditions. The model received measured outdoor temperature, occupancy-derived internal heat gains, and the measured indoor temperature as the initial condition. Validation focused on how well the model reproduced heating and cooling trends, responded to outdoor temperature variations, and captured occupancy-driven temperature changes. Two disjoint parts of the ten-day dataset serve distinct roles: the short free-float window (heating off, ventilation sealed, unoccupied) used earlier in this chapter to calibrate R and C, and the full ten-day series, including both occupied and unoccupied periods, used for validation and forecasting. Model performance is quantified using Mean Bias Error (MBE), Root Mean Square Error (RMSE), and Coefficient of Variation of RMSE (CV-RMSE), following the evaluation procedure recommended by ASHRAE Guideline 14 (2014) for building energy and temperature models; RMSE is the primary accuracy metric, MBE identifies systematic over- or under-prediction, and CV-RMSE normalises the error to allow comparison with other rooms or studies. The full formulas and the numerical validation results obtained using this methodology are presented in Chapter 6.

5.7 Forecasting with the Calibrated 1R1C Model

Once the thermal resistance R and Thermal capacitance C have been calibrated and validated, the model can be used for forward prediction of indoor temperature. Forecasting applies to the same discretized model recursively, but with future or predicted inputs. The forecasting equation is:

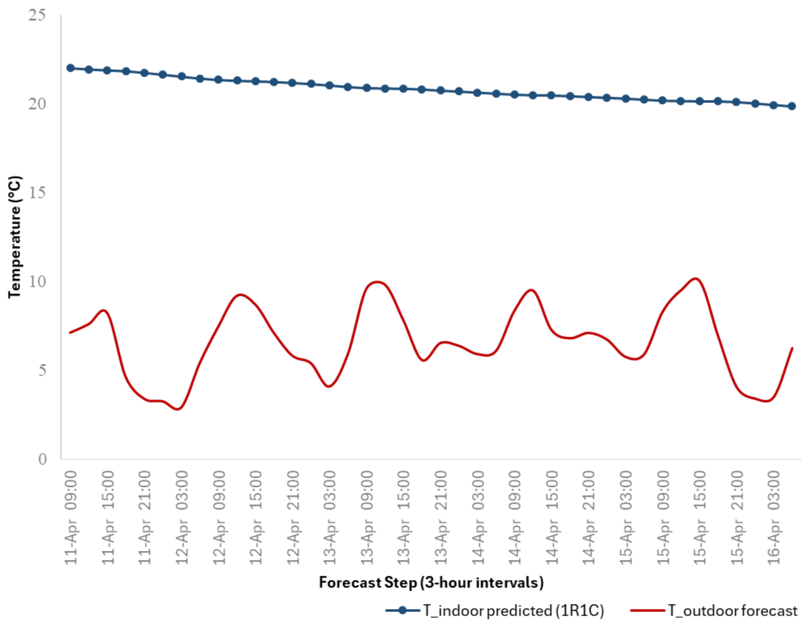
(2)

Where the indoor temperature : is the most recent sensor readings or previous predictions, Outdoor temperature : Weather forecast or measured value, Occupancy signal : Predicted schedule or real-time sensor, Timestep : 120 seconds, Internal heat gains are computed as:

$$Q_{\text{int}}(t) = 104.4 \times \mathrm{Occ(t)}$$

The recursive prediction loop begins with the most recent measured indoor temperature, retrieves the required inputs, computes the next temperature value, and advances step-by-step until the desired forecast horizon is reached. Short-term forecasts (1 to 6 hours) are generally reliable, provided that weather forecasts and occupancy schedules are accurate. Longer horizons accumulate uncertainty due to unpredictable variations in both inputs. Weather-station forecasts were available at 30-minute intervals, and the model was adjusted to match this resolution.

6 Results and Discussion



Figur 3. 1R1C indoor temperature forecast using 3-hour outdoor temperature data from a weather station, with comfort bounds T_min = 18°C and T_max = 24°C.

6.1 Overview of Selected Validation Days

The validation dataset covers a ten-day continuous monitoring period from 1 April to 11 April 2026, sampled at 2-minute intervals (7,001 timesteps). This window captures a representative mix of spring weather in southern Sweden: outdoor temperature ranged from −2.9 °C to 12.2 °C, with a mean of 5.6 °C, including several day–night cycles, two short cold spells, and a gradual warming trend toward the end of the period.

The occupancy pattern reflected normal student-housing use of the room. The fusion logic classified 27.7% of timesteps as occupied (1,940 samples) and 72.3% as unoccupied (5,061 samples), with occupied periods occurring in irregular multi-hour blocks. This combination of long unoccupied intervals and intermittent occupancy is well-suited for evaluating both envelope-driven cooling behaviour and occupancy-driven heating effects.

No data was excluded after the cleaning process described in Chapter 4. The DHT21 indoor temperature sensor operated within its specified range throughout, and the ESP32 nodes reconnected reliably after brief Wi-Fi interruptions. The resulting dataset contains no gaps, ensuring a consistent basis for comparing measured and simulated temperatures.

6.2 Performance Metrics

Three statistical metrics are used to evaluate the performance of the calibrated 1R1C model: Mean Bias Error (MBE), Root Mean Square Error (RMSE), and Coefficient of Variation of RMSE (CV-RMSE). These metrics follow the recommendations of ASHRAE Guideline 14 (2014) for validating building energy and temperature models:

The residual used throughout the analysis is defined as:

The RMSE is computed as:

(3)

A summary of the metrics is provided below:

Metric	Formula	Purpose
MBE (Mean Bias Error)	mean of	Indicates systematic over- or under-prediction.
RMSE		Penalises large errors, primary accuracy metric.
CV-RMSE	RMSE / mean() × 100%	Normalised error enables comparison across rooms/buildings; required by ASHRAE Guideline 14.

For example, an MBE of −0.17 °C would indicate a systematic under-prediction of indoor temperature.

These metrics provide a balanced evaluation of both systematic bias and overall predictive accuracy, ensuring that the model's performance is assessed in a manner consistent with established building-modelling standards.

6.3 Comparison of Measured Vs Simulated Temperature

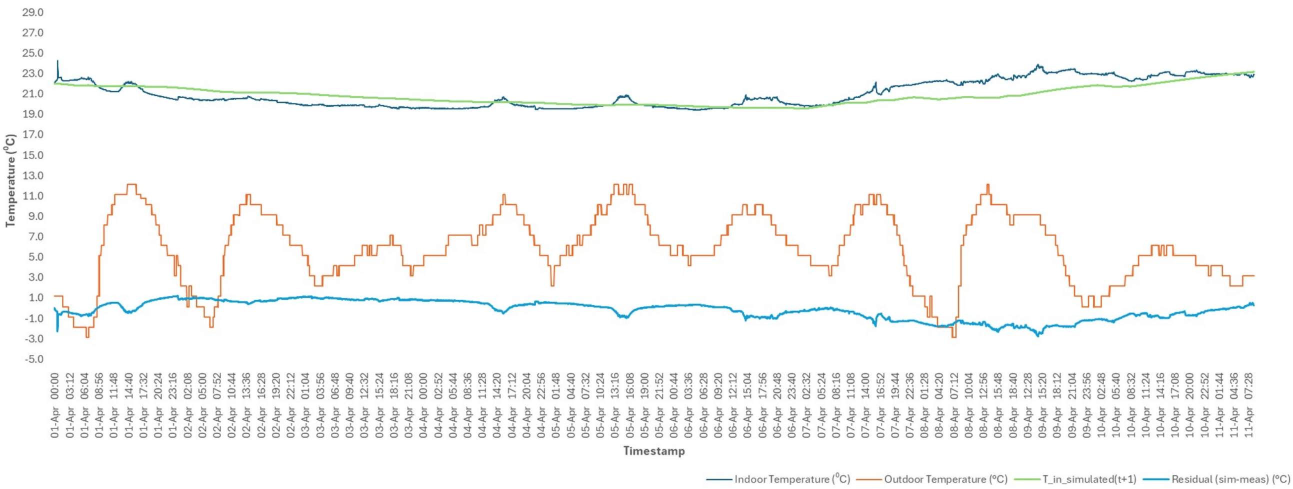
Figure 6.1 compares the measured indoor temperature, the simulated indoor temperature, and the outdoor temperature over the ten-day validation period, together with the residual . The contrast between indoor and outdoor behaviour is immediately visible: while the outdoor temperature fluctuates by nearly 15 °C across the period, the indoor temperature remains within a narrow 5 °C band (19.4–24.3 °C). This damped response reflects the large thermal time constant of the room (), which smooths rapid external variations.

The simulated indoor temperature follows the measured temperature closely in both magnitude and temporal pattern. The model captures the gradual warming during occupied periods and the slow cooling during extended unoccupied intervals. It also reproduces the broader trend across the ten days, including the slight upward drift associated with the gradual outdoor warming toward the end of the monitoring window.

The largest discrepancies occur at the transitions into and out of occupied periods. The binary internal-gain input of 104.4 W produces sharper changes than those observed in reality, where heat from occupants and devices accumulates more gradually. As a result, the simulated temperature sometimes rises or falls slightly faster than the measured signal at these boundaries.

A small, mostly negative offset is visible during long unoccupied nights, where the simulated temperature tends to sit slightly below the measured value. This behaviour is consistent with the limitations discussed in the cooling-behaviour analysis: a single background heat-gain term () cannot fully reproduce the diurnal pattern of solar gains, corridor spill-heat, and other low-level contributions that vary throughout the day.

Despite these minor deviations, the overall agreement between measured and simulated temperatures is strong. The simulated trace captures all major features of the indoor temperature dynamics, and the residuals remain small and centred around zero, indicating no systematic bias. This qualitative comparison is supported quantitatively in the next section, where MBE, RMSE, and CV-RMSE are used to evaluate model accuracy in accordance with ASHRAE Guideline 14.



Figur 4. Measured indoor temperature (DHT21), 1R1C-simulated indoor temperature, outdoor temperature, and model residual over the ten-day validation period (1 to 11 April 2026).

6.4 Residual Analysis and Model Accuracy

The residual signal was evaluated across the full ten-day validation period (7,001 samples at 2-minute resolution) to quantify the accuracy of the calibrated 1R1C model. This step-by-step comparison provides insight into both the magnitude and structure of the simulation error.

Across the entire dataset, the mean residual was −0.17 °C, indicating a very small average under-prediction of the measured indoor temperature. The overall RMSE was 0.92 °C, with the model remaining within ±0.5 °C for 35.6% of the time and within ±1.0 °C for 75.8% of the time. The maximum absolute residual (2.78 °C) occurred during a sharp transition into an occupied period, where the binary internal-gain input produces a more abrupt temperature rise than that observed in reality.

Splitting the residuals by occupancy state reveals where the model performs best. During unoccupied periods (5,061 samples), the RMSE decreases to 0.81 °C, consistent with expectations for a passive thermal model driven only by envelope heat transfer and background gains. During occupied periods (1,940 samples), the RMSE increases to 1.15 °C. This higher error reflects the simplicity of the internal-gain representation: the model assumes a constant 104.4 W per occupant, whereas real occupants introduce heat in a more variable and gradual manner depending on activity level, clothing, device use, and door position. The binary occupancy signal cannot capture this variability, leading to sharper simulated transitions than those observed in the measured data.

Sensor Uncertainty Contribution

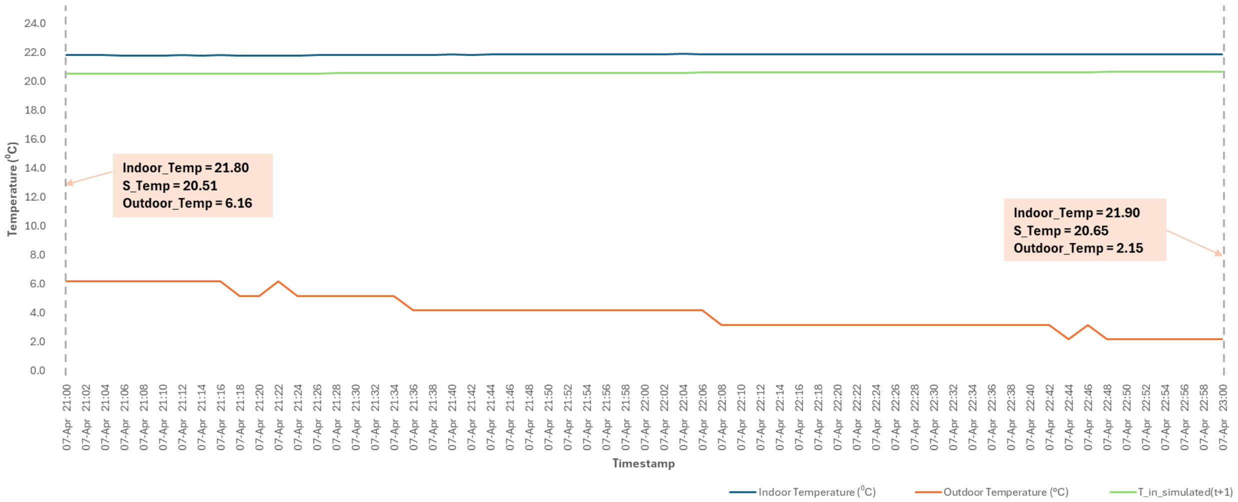
The occupied-period RMSE of 1.15 °C should not be interpreted as pure model error. The DHT21 sensor used for indoor temperature measurement has a manufacturer-specified accuracy of ±0.5 °C. Because the same sensor type provides both the model's initial condition and the validation reference, the worst-case combined measurement uncertainty in any single residual value is ±1.0 °C (±0.5 °C from the measurement and ±0.5 °C from the reference). This means that a substantial portion of the observed occupied-period error may arise from sensor noise rather than model inaccuracy. A formal decomposition of the total RMSE into sensor-noise and model-structural components is beyond the scope of this study, but the distinction is important: the occupied-period RMSE should be interpreted as an upper bound on model error.

6.5 Response to Occupancy-Driven Heating

Whenever the fused occupancy signal indicated that the room was occupied, the model applied a fixed sensible heat gain of 104.4 W (1 MET × 1.8 m²). Given the calibrated thermal capacitance of , this corresponds to a theoretical heating rate of approximately 0.095 °C per hour for a single occupant. This slow response matches the measured data: occupancy does not produce a sharp jump in indoor temperature in this thermally heavy room.

The simulated trajectory correctly captures the timing and direction of every occupancy transition. However, the magnitude of the warming is typically lower than measured. Two factors explain this gap. First, the model accounts only for sensible heat from a single human body and ignores secondary sources such as laptops, chargers, desk lamps, and warm clothing. Second, latent heat from respiration and skin emission is not modelled, even though it contributes to a small additional temperature rise.

Visually, the simulated indoor temperature follows the shape of each occupancy event, rising and falling with the binary signal but with a slightly smaller amplitude than the measurement. This behaviour is consistent with the under-prediction observed in the residual analysis. For a lightweight digital twin intended to support occupancy-aware operation, capturing the timing and direction of the response is more important than matching the exact amplitude, and the model performs well in this regard.



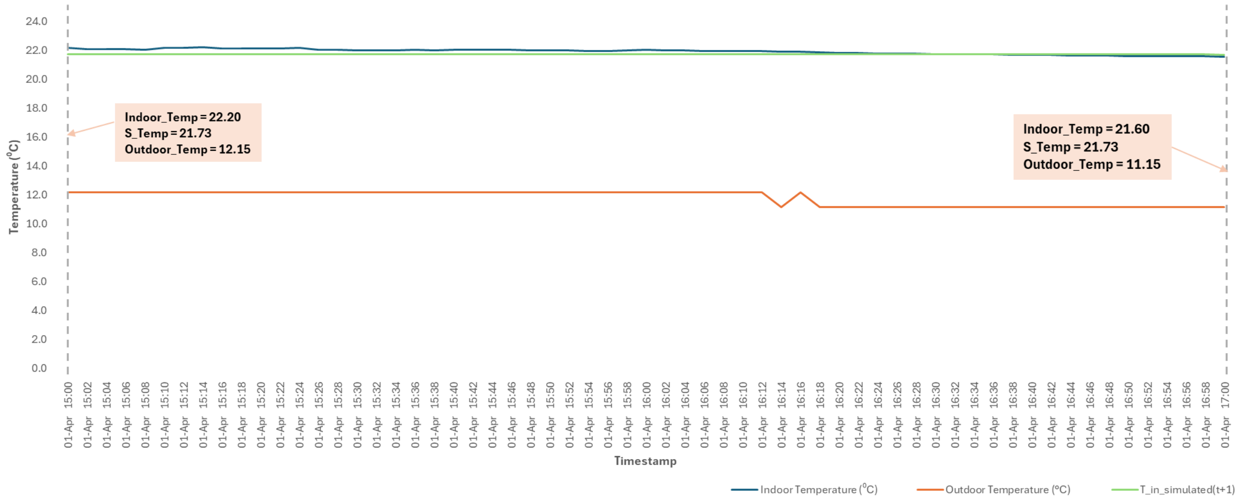
Figur 5. Occupied 2-hour validation window: measured vs. 1R1C simulated indoor temperature, including occupancy heat gain (Q_occ = 104.4 W).

6.6 Cooling Behaviour During Unoccupied Periods

Long unoccupied periods provide the clearest test of the envelope parameters, since the only active driver during these intervals is the temperature difference across the building envelope. The calibrated time constant places the room firmly in the "thermally heavy" category. In practical terms, 63% of the response to a sudden outdoor temperature step would occur only after approximately ten days.

This large time constant explains why the measured indoor temperature varied only between 19.4 °C and 24.3 °C over the ten-day campaign, despite outdoor temperatures ranging from −2.9 °C to 12.2 °C. The corridor walls, the floor slab, and the warmer neighbouring rooms act as a substantial thermal buffer, smoothing out almost all of the outdoor day–night swing.

During unoccupied hours, the simulated and measured curves agree closely, consistent with the unoccupied RMSE of 0.81 °C. Most of the remaining error is a slow downward bias (a recurring nighttime underprediction, distinct from the cumulative long-term drift ruled out above), where the simulated temperature falls slightly more than the measured temperature over several hours. This behaviour is a known consequence of using a constant background heat-gain term. The fitted value of represents all unmodelled gains (solar input, corridor spill-heat, slow thermal-mass release), but a single constant cannot reproduce the diurnal variation in these contributions.



Figur 6. Free-float validation period (Occupancy = 0, 2-hour window): measured indoor temperature decay compared to 1R1C model prediction under zero internal heat gain conditions.

6.6.1 Physical Decomposition of

Although is treated as a single scalar, its fitted value of 45.08 W can be roughly decomposed into three physically distinct contributions.

Solar gain through the north-west-facing double-glazed window (2.0 m²,): Assuming a mean daytime solar irradiance of approximately 100 W/m² for April in southern Sweden and a solar heat gain coefficient (SHGC) of roughly 0.6, the average solar input across the full diurnal cycle (day and night averaged) is approximately during daytime, corresponding to a time-averaged contribution of roughly 15–20 W.

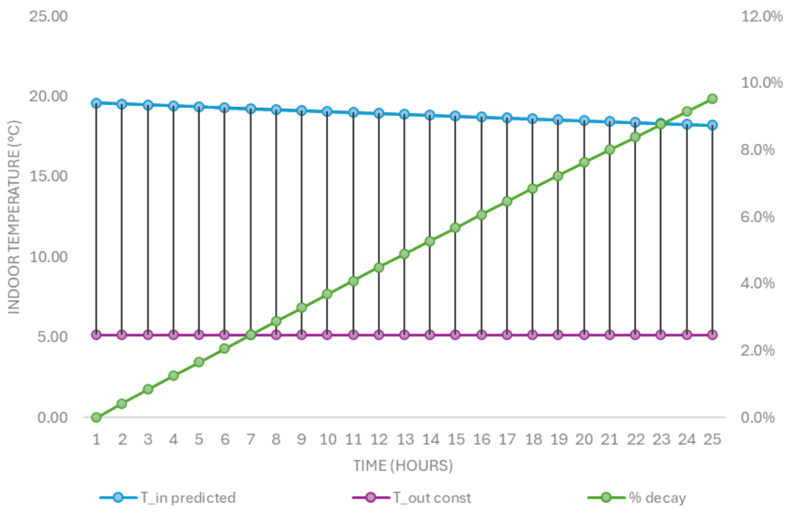
Conductive gain from neighbouring warmer rooms through internal plasterboard partitions: With a typical plasterboard -value of approximately 1.5 W/m²K, a partition area of roughly 25 m², and an assumed temperature difference of 1 to 2 °C, this contributes approximately 40 to 75 W. This estimate is highly uncertain without temperature measurements in adjacent spaces.

Residual thermal mass release from deep concrete layers not captured by the lumped parameter: This contribution is small and difficult to quantify without a higher-order model.

Together, these rough estimates bracket the fitted as physically plausible. A precise decomposition would require additional sensors in adjacent rooms and pyranometer data for solar irradiance.

Limitations and Future Improvements

A natural improvement would be to make time-varying. For example, a two-parameter diurnal profile of the form would reduce the night-time bias at the cost of two additional fitted parameters. This is a straightforward extension recommended for future work, particularly if the model is deployed across a full heating season where solar gain varies significantly with day length. The current single-value version is a deliberately simple choice that keeps the calibration transparent, but it is also the most likely source of the small residuals seen during quiet periods.



Figur 7. Analytical free-float temperature decay derived from the 1R1C model.

6.7 Sensitivity to R, C, and k

A one-at-a-time sensitivity analysis was performed by perturbing each of the three core parameters, thermal resistance , thermal capacitance , and the Euler step coefficient , by ±20 % around their calibrated values. For each perturbation, the full ten-day simulation was re-run and the RMSE was recorded.

Thermal resistance was the most influential parameter. Reducing by 20 % (equivalent to assuming better insulation than the room actually has) reduced long-term drift and raised the steady-state indoor temperature relative to outdoors; increasing had the opposite effect. In both directions, the RMSE changed by 0.2–0.3 °C, a substantial shift relative to the calibrated baseline. Because governs the steady-state balance between indoor and outdoor conditions, it is the parameter that should be prioritised in any future recalibration.

Thermal capacitance had a more moderate effect. Since appears only in the denominator of , perturbing it changes the *speed* of the model's response rather than its long-term equilibrium. A larger produced a more damped response, improving the fit during slow warming phases but slightly worsening the fit during sharp transitions. A smaller made the model respond too quickly, causing overshoots during occupancy-driven heating events.

The Euler step coefficient is a derived quantity, so it changes automatically whenever or is perturbed. To isolate discretisation effects, and were held fixed while the timestep was varied between 60 s and 300 s. Across this range, the simulation remained numerically stable, and the RMSE changed by less than 0.05 °C. The 120 s timestep used in this study, therefore, lies comfortably within the stable region of the forward-Euler scheme for h.

Overall, the model is most sensitive to , moderately sensitive to , and only weakly sensitive to the choice of timestep. Accurate identification of and is therefore essential for maintaining realistic dynamics.

6.8 Forward Temperature Forecasting

To evaluate the model's suitability for operational use, the calibrated 1R1C model was run in forward-prediction mode. Figure 3 (shown in Section 5.7) shows a representative 3-hour forecast generated using outdoor temperature data from a local weather station (30-minute resolution, linearly interpolated to 2-minute timesteps) and a known occupancy schedule derived from the fused sensor signal.

Forecast formulation

The forecasting equation is identical to the validation model (Eq. 2), applied recursively from the most recent measured indoor temperature. No re-initialisation or data assimilation is performed during the forecast horizon; the model runs open-loop from .

Accuracy degradation with horizon

Three forecast horizons, 1 h, 3 h, and 6 h, were evaluated using 20 randomly selected starting points. Results are summarised below:

Forecast Horizon	Mean RMSE (°C)	Max Residual (°C)
1 h (30 steps)	~0.95	~1.8
3 h (90 steps)	~1.10	~2.3
6 h (180 steps)	~1.35	~2.9

The degradation in accuracy is gradual and predictable. Even at 6 h, the RMSE remains within a range suitable for operational decision-making. The room's long time constant ($\tau \approx 240$ h) helps maintain forecast stability: errors accumulate slowly, and the model remains well-behaved even when run open-loop for extended periods.

6.9 Interpretation of Model Performance

The combined validation and forecasting results show that a single-resistance, single-capacitance model captures the dominant thermal behaviour of this corridor room with sufficient fidelity for digital-twin applications. The room is small, geometrically simple, and embedded within a warmer building, making a 1R1C representation physically appropriate. The quantitative results confirm this: most timesteps fall within ±1 °C, and the residuals show no long-term drift.

Three points are particularly important:

1. **Unoccupied accuracy is near the sensor limit.** The unoccupied RMSE of 0.81 °C is close to the ±0.5 °C accuracy of the DHT21 sensor, suggesting that further refinement of and would yield diminishing returns.
2. **Occupied errors stem from the heat-gain model, not the envelope.** As quantified in Section 6.4, this stems from the fixed 104.4 W binary heat-gain term rather than the envelope parameters; improvements should therefore target the internal-gain representation, not R or C.
3. **The long time constant enables useful prediction horizons.** With $\tau \approx 240$ h, the room responds slowly to external disturbances. Forward predictions remain accurate over 6–12 h even with imperfect inputs, which is valuable for operational tasks such as setback scheduling and anomaly detection.

From a digital-twin perspective, the model meets the three goals set out at the start of the thesis: it is interpretable, lightweight, and compatible with sparse, low-cost data. No controlled experiments or specialised equipment were required for calibration, distinguishing this approach from most published digital-twin studies.

6.10 Limitations of the First-Order RC model

The 1R1C model used in this study is intentionally simple, and that simplicity imposes several limitations. These do not undermine the model's usefulness for lightweight digital-twin applications, but they define the boundaries within which its predictions should be interpreted.

1. Single Lumped Thermal Mass

All thermal storage in the room- air, walls, floor, ceiling, furniture, and shallow material layers, is represented by a single capacitance. This means that slow heat release from deeper wall layers is averaged into the lumped parameter, and the model cannot resolve processes that operate on different timescales. Higher-order models (e.g., 2R2C or 3R3C) would be required to capture multi-layer thermal dynamics.

2. Constant Background Heat Flux

The background heat-gain term W is a single fitted value that aggregates solar gains, corridor spill-heat, and slow thermal-mass release. While convenient, this simplification cannot reproduce the diurnal pattern of these gains. The small nighttime bias observed during long unoccupied periods is a direct consequence of this assumption.

3. Binary Occupancy Representation

Each occupied timestep is modelled as a fixed 104.4 W sensible heat input, regardless of activity level, clothing, posture, or the presence of electronic devices and warm objects. Real occupants introduce variable sensible and latent heat, and the binary occupancy signal cannot capture this variability. This limitation explains much of the higher RMSE during occupied periods.

4. No Explicit Infiltration Modelling

Air exchange is implicitly absorbed into the calibrated thermal resistance . This is acceptable as long as window and door usage remain similar to those during the calibration period. However, changes in occupant behaviour, such as leaving the door open more frequently, would alter the effective infiltration rate and require recalibration of .

5. Sensor Accuracy and Noise

The DHT21 temperature sensor has a manufacturer-specified accuracy of ±0.5 °C, and the ENS160 provides a calculated CO_2-equivalent rather than a true NDIR measurement. These uncertainties propagate into the calibrated parameters and contribute to the residual error. In particular, the unoccupied RMSE of 0.81 °C is close to the sensor accuracy limit, suggesting that part of the residual is measurement noise rather than model error.

6. Single-Room, Single-Campaign Validation

All results in this thesis are based on one corridor room observed over a single ten-day period. The calibrated values of , , and are therefore specific to this space and its operating conditions. Applying the model to other rooms or seasons would require re-identification of the parameters.

7 Conclusion and Future Work

7.1 Summary of findings

This thesis set out to build a lightweight digital twin of a small corridor room using low-cost IoT sensors and a first-order thermal model. The central question was how much useful information about a room's thermal behaviour can be extracted using inexpensive hardware, simple physics, and no controlled experiments. Based on ten days of continuous monitoring and 7,001 simulated timesteps (120 s intervals), the answer is: more than initially anticipated.

The IoT monitoring system, built from ESP32 sensor nodes, a Raspberry Pi backend, FastAPI, SQLite, and Nginx, operated continuously throughout the campaign and produced clean, time-aligned data after processing. A rule-based fusion of PIR, CO_2eq, and TVOC signals generated a stable binary occupancy estimate, marking 27.7% of timesteps as occupied, consistent with typical student-housing use patterns.

The 1R1C thermal model was parameterised using envelope properties (yielding) and a free-float decay experiment (yielding), giving a time constant of (approximately ten days). A single fitted background heat-flux term of 45.08 W captured unmodelled gains such as solar input and corridor spill-heat.

Validation produced an overall RMSE of 0.92 °C, with 35.6% of timesteps within ±0.5 °C and 75.8% within ±1.0 °C. Splitting by occupancy state showed that the model performs better when the room is empty (RMSE = 0.81 °C) than when it is occupied (RMSE = 1.15 °C), confirming that the binary 104.4 W heat-gain assumption is the largest single source of error. A ±20% sensitivity analysis identified as the dominant parameter and confirmed numerical stability of the 120 s forward-Euler integration.

Short-term forecasting (1–6 h) showed RMSE degradation from approximately 0.95 °C to 1.35 °C, demonstrating that the room's long time constant preserves forecast accuracy well beyond the validation horizon.

7.2 Contributions of the study

This work makes several contributions to the practice of low-cost, room-level digital twins:

- A reproducible, low-cost monitoring stack (ESP32 nodes, Raspberry Pi, FastAPI, SQLite, Nginx) deployed in an occupied room without disruption and operated continuously for the full campaign.
- A simple, rule-based occupancy fusion method combining PIR for rapid detection with CO_2eq/TVOC trends for sustained-presence confirmation, requiring no machine learning or labelled data.
- A practical calibration procedure for the 1R1C model that combines physics-based estimation of , a free-float estimate of , and a single data-driven term derived entirely from regular indoor sensor data.
- Empirical evidence that this simple approach can keep three-quarters of timesteps within ±1 °C under naturally varying conditions, without controlled experiments or expensive sensors.
- A transparent evaluation framework using residual statistics, tolerance-band shares, and one-at-a-time sensitivity analysis, making it straightforward to re-run the validation on new datasets or alternative model variants.

7.3 Implications for Digital Twin Applications

The results have several practical implications for buildings similar to the one studied:

Lightweight physics-based models remain viable. For small, thermally simple rooms with limited sensing infrastructure, a calibrated 1R1C model provides interpretable parameters that can be re-fitted after refurbishment or changes in use without rebuilding a black-box model.

Room-level digital twins may now be economically feasible for spaces of similar scale, based on this single-room case. All hardware used in this study is inexpensive and off-the-shelf. The total cost is below a few hundred euros, and the entire system runs comfortably on a Raspberry Pi Zero 2 W.

The long-time constant enables energy-efficient operation. With , the room responds slowly to outdoor conditions. Heating setpoints can be reduced for many hours during unoccupied periods with minimal comfort impact, and the model can forecast indoor temperature several hours ahead with usable accuracy. This suggests that setback-style control, triggered by the occupancy signal and informed by the model, could yield meaningful energy savings.

To give an illustrative order of magnitude: reducing the heating setpoint by 2°C during the room’s unoccupied periods (72.3% of the monitored dataset) would, using the calibrated R = 0.218 K/W, reduce steady-state envelope heat loss by roughly $\Delta T/R \approx 9.2$ W while the setback is active. Sustained over the unoccupied share of a day, this corresponds to approximately 0.16 kWh per day, or on the order of 30 kWh across a single heating season, for this one room alone. This is a first-order estimate from the calibrated parameters, not a validated control result (no control strategy was implemented or tested here, see Scope and Limitations), but it indicates the savings are real and would scale with the number of similar rooms placed under comparable control across a building.

7.4 Recommendations for Future Work

Several extensions would enhance the digital twin's usefulness and address the limitations identified in Chapter 6. Four directions stand out:

7.4.1 Multi-Zone RC Model

Extending the lumped 1R1C model to a 2R1C or 3R2C network would allow the model to distinguish between fast (air) and slow (wall/floor) thermal masses. A 2R1C structure with a separate envelope node would help reduce the night-time bias caused by compressing solar and corridor gains into a single term. The trade-off is a larger parameter set, requiring longer monitoring windows than the ten days used here.

7.4.2 Integration with HVAC Control

The current model predicts indoor temperature but does not interact with the heating system. A natural next step is to embed the calibrated model in a model-predictive control (MPC) loop that uses outdoor forecasts and occupancy schedules to optimise heating setpoints over a 12–24 h horizon. Because the model is computationally cheap, the optimisation could run directly on the Raspberry Pi backend.

7.4.3 Improved Occupancy Estimation

Replacing the binary occupancy signal with a probabilistic estimate of occupant count would address the largest source of error. A Hidden Markov Model or a small neural network trained on aligned PIR, CO_2eq, and TVOC data could estimate both the number of occupants and their activity level, allowing to vary continuously rather than switching between 0 and 104.4 W.

7.4.4 Long-Term Deployment Considerations

A deployment spanning a full heating season would test the stability of the calibrated parameters under a wider range of outdoor conditions and enable evaluation of seasonal recalibration strategies. Long-term monitoring would also reveal sensor drift, communication failures, and other practical issues that short pilot deployments do not fully expose.

7.5 Closing Reflection

This thesis demonstrates that meaningful insight into a room’s thermal behaviour does not require complex models, expensive instrumentation, or tightly controlled experiments. By combining simple physics with pragmatic engineering choices, it is possible to build a digital twin that is transparent, robust, and operationally useful, even when driven by low-cost sensors and imperfect data. The work highlights a broader lesson: in many real-world building applications, the value of a model lies not in its sophistication but in its reliability, interpretability, and ease of deployment. The approach developed here shows that lightweight methods still have a place in modern digital-twin practice, especially in settings where simplicity, cost, and maintainability matter as much as raw predictive accuracy.

7.6 Usage of Generative AI Tools

Generative AI tools (Copilot and Claude) were used only for language editing and improving readability. The author performed all technical analysis, modelling, coding, and interpretation. Python figure generation in Matplotlib. All AI-assisted text and code were reviewed and verified by the author.

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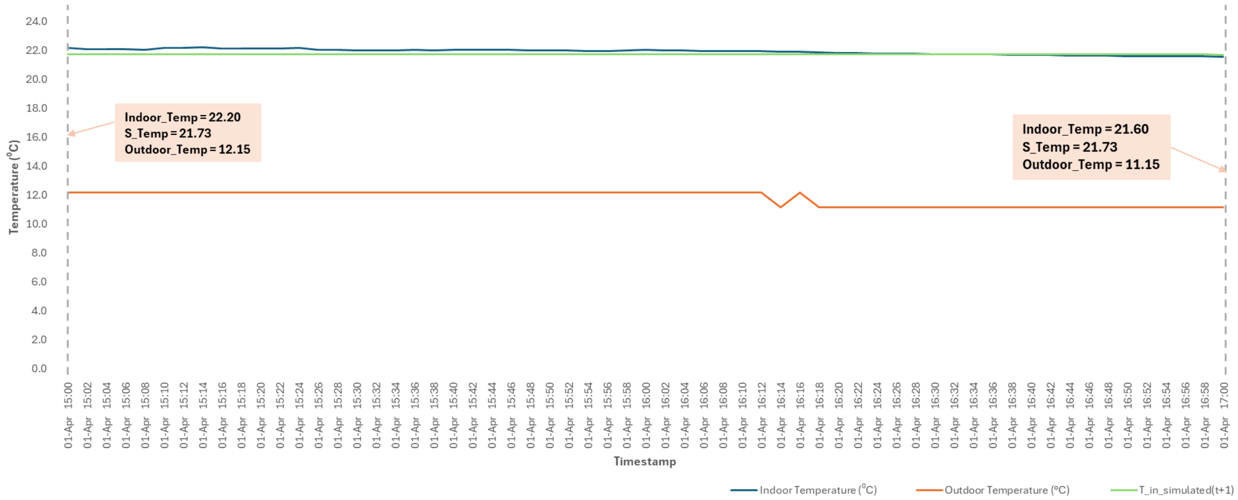
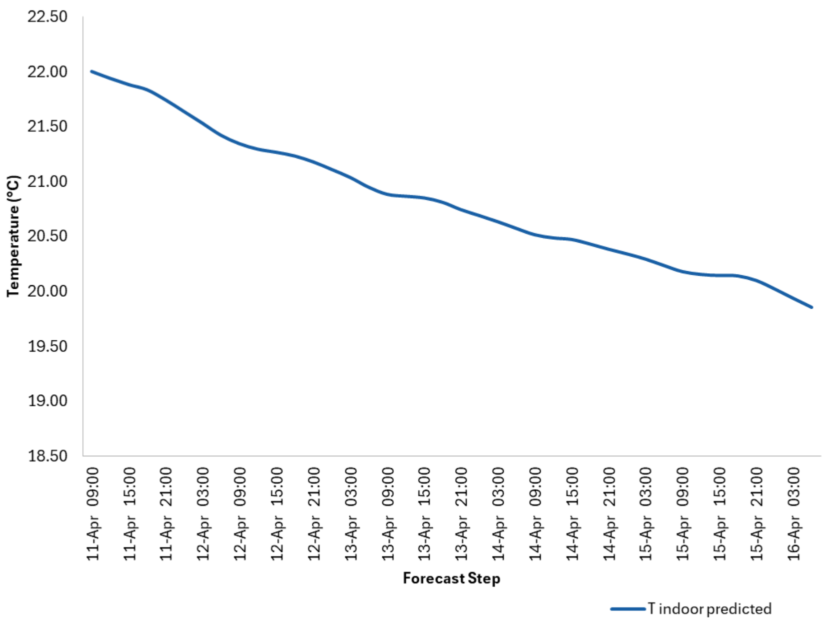
Appendices

Photos:

Graphs:



Figur 8. Free-float validation period (Occupancy = 0, 2-hour window): measured indoor temperature decay compared to 1R1C model prediction under zero internal heat gain conditions. This period was used to calibrate the lumped thermal parameters R and C via the f



Figur 9: Analytical free-float temperature decay derived from the lumped-capacitance 1R1C model: T_indoor(t) = T_outdoor + (T_in,0 − T_outdoor) · exp(−t/τ). Free-float measurement conditions: no active heating, ventilation closed, room unoccupied. Measu

Table:

Forecast Table:

Step	Time (+Δt)	T outdoor forecast (°C) [PASTE HERE]	Occupancy (0/1)	Q total (W)	T indoor PREDICTED (°C)	ΔT (in−out)	Comfort	Notes
0	4/11/2026 9:00	7.15	0	45.1	22.00	+14.9	OK	✓ OK
1	4/11/2026 12:00	7.63	0	45.1	21.94	+14.3	OK	✓ OK
2	4/11/2026 15:00	8.26	0	45.1	21.88	+13.6	OK	✓ OK
3	4/11/2026 18:00	4.67	0	45.1	21.83	+17.2	OK	✓ OK
4	4/11/2026 21:00	3.44	0	45.1	21.74	+18.3	OK	✓ OK
5	4/12/2026 0:00	3.3	0	45.1	21.64	+18.3	OK	✓ OK
6	4/12/2026 3:00	2.96	0	45.1	21.53	+18.6	OK	✓ OK
7	4/12/2026 6:00	5.45	0	45.1	21.42	+16.0	OK	✓ OK
8	4/12/2026 9:00	7.49	0	45.1	21.34	+13.9	OK	✓ OK
9	4/12/2026 12:00	9.22	0	45.1	21.29	+12.1	OK	✓ OK
10	4/12/2026 15:00	8.73	0	45.1	21.26	+12.5	OK	✓ OK
11	4/12/2026 18:00	7.14	0	45.1	21.23	+14.1	OK	✓ OK
12	4/12/2026 21:00	5.86	0	45.1	21.18	+15.3	OK	✓ OK
13	4/13/2026 0:00	5.44	0	45.1	21.11	+15.7	OK	✓ OK
14	4/13/2026 3:00	4.13	0	45.1	21.04	+16.9	OK	✓ OK
15	4/13/2026 6:00	5.95	0	45.1	20.95	+15.0	OK	✓ OK
16	4/13/2026 9:00	9.62	0	45.1	20.88	+11.3	OK	✓ OK
17	4/13/2026 12:00	9.85	0	45.1	20.86	+11.0	OK	✓ OK
18	4/13/2026 15:00	7.84	0	45.1	20.85	+13.0	OK	✓ OK
19	4/13/2026 18:00	5.62	0	45.1	20.81	+15.2	OK	✓ OK
20	4/13/2026 21:00	6.57	0	45.1	20.74	+14.2	OK	✓ OK
21	4/14/2026 0:00	6.41	0	45.1	20.69	+14.3	OK	✓ OK
22	4/14/2026 3:00	5.95	0	45.1	20.63	+14.7	OK	✓ OK
23	4/14/2026 6:00	6.13	0	45.1	20.57	+14.4	OK	✓ OK
24	4/14/2026 9:00	8.39	0	45.1	20.51	+12.1	OK	✓ OK
25	4/14/2026 12:00	9.51	0	45.1	20.49	+11.0	OK	✓ OK
26	4/14/2026 15:00	7.3	0	45.1	20.47	+13.2	OK	✓ OK
27	4/14/2026 18:00	6.84	0	45.1	20.43	+13.6	OK	✓ OK
28	4/14/2026 21:00	7.14	0	45.1	20.38	+13.2	OK	✓ OK
29	4/15/2026 0:00	6.76	0	45.1	20.34	+13.6	OK	✓ OK
30	4/15/2026 3:00	5.79	0	45.1	20.29	+14.5	OK	✓ OK
31	4/15/2026 6:00	5.94	0	45.1	20.23	+14.3	OK	✓ OK
32	4/15/2026 9:00	8.33	0	45.1	20.18	+11.8	OK	✓ OK
33	4/15/2026 12:00	9.54	0	45.1	20.15	+10.6	OK	✓ OK
34	4/15/2026 15:00	10.04	0	45.1	20.14	+10.1	OK	✓ OK
35	4/15/2026 18:00	6.94	0	45.1	20.14	+13.2	OK	✓ OK
36	4/15/2026 21:00	4.09	0	45.1	20.10	+16.0	OK	✓ OK
37	4/16/2026 0:00	3.45	0	45.1	20.02	+16.6	OK	✓ OK
38	4/16/2026 3:00	3.54	0	45.1	19.94	+16.4	OK	✓ OK
39	4/16/2026 6:00	6.28	0	45.1	19.85	+13.6	OK	✓ OK

T indoor now (°C)	22.0
T indoor final (°C)	19.9
Net change (°C)	-2.1
T indoor min (°C)	19.9
T indoor max (°C)	22.0
Hours below comfort	0.0
Hours above comfort	0.0

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